Minimum-Effort Planning for Constant-Power Spacecraft Timing, Engine Limits and Planetary Encounters

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Abstract

Constant useful jet power turns propellant expenditure into an integral of squared thrust acceleration. This gives a common basis for transfer planning, local rendezvous and planetary encounters. We develop a fixed-time minimum-effort formulation, derive moving-endpoint timing gradients, and show how to invert an effort budget to search for minimum flight time. Finite exhaust speed makes the admissible acceleration depend on mass; the resulting resource adjoint, hard radial constraints and moving-interface conditions are incorporated in the same convention. Exact linear rendezvous results and a globally solved circular-spiral approximation provide inexpensive local calculations. For full Kepler transfers, second variations and global effort bounds separate stationary branches from verified performance. Position-only interception and deadline-fuel games are distinguished from full-state rendezvous. Reproducible checks include nonconvex timing, capped free-space transcription, checks of earlier Jupiter-class candidates, and an exact local solar path with global effort and time bounds. An explicit band-feasible Earth–Mars circular-orbit witness also supplies a certified 70-day flight-time upper bound for its specified departure phase. The resulting framework specifies the mathematical and computational interfaces for a planner; a globally validated station-to-station trajectory atlas remains to be constructed.

1 From propellant to a planning problem

The planning task is to move a spacecraft between prescribed orbital states, possibly through planetary encounters, with finite propellant and constant available useful jet power. The central question is how much effort a specified arrival date requires. Once that function is available, the earliest affordable arrival follows from a time search. This order of calculation also exposes the marginal cost of changing dates and the conditions needed to join independently represented trajectory phases.

All motion here is planar. Planet ephemerides are prescribed circular, coplanar orbits; stations may orbit a planet or moon. Within each gravitational patch the relative dynamics are controlled Kepler motion. An unpowered point flyby and a finite planetary segment are alternative encounter models. The ideal engine supplies useful analytic benchmarks; the operational model uses full power or coast with exhaust speed in the specified 50–250 km/s band. Observations in the pursuit problem are instantaneous.

1.1 The exact mass identity

Let $P > 0$ be constant useful jet power during a burn, m the spacecraft mass, v_e exhaust speed and a thrust acceleration magnitude. Then

$$P = \tfrac{1}{2}(-\dot m)v_e^2, \qquad ma = (-\dot m)v_e. \tag{1}$$

Eliminating v_e gives

$$\begin{gathered}\frac{d}{dt}\frac{1}{m} = \frac{a^2}{2P}, \qquad j = \int_0^T a^2\,dt, \\ m_f = \left(\frac{1}{m_0} + \frac{j}{2P}\right)^{-1}.\end{gathered} \tag{2}$$

Coasting has $a = 0$ and constant mass. Thus, for fixed initial mass and power, minimizing propellant is exactly minimizing j. With dry mass m_d, the available physical effort is $B = 2P(m_d^{-1} - m_0^{-1})$. Electrical input power must first be converted to useful jet power. The assumed P does not increase as a fixed spacecraft becomes lighter.

For a primary with parameter μ, choose a length R_0 and define

$$\begin{aligned} T_0 &= \sqrt{R_0^3/\mu}, \quad V_0 = \sqrt{\mu/R_0}, \\ A_0 &= \mu/R_0^2, \quad J_0 = V_0^2/T_0. \end{aligned} \tag{3}$$

Dimensionless variables then obey

$$\begin{gathered} r' = v, \qquad v' = g(r) + u, \\ g(r) = -r/|r|^3, \qquad J = \int_0^h |u|^2\,dt, \end{gathered} \tag{4}$$

where $h = T/T_0$ and $j = J_0 J$. Define

$$\begin{gathered} \chi = \frac{2P}{m_0 J_0}, \quad \chi_d = \frac{2P}{m_d J_0}, \quad \kappa = \chi_d - \chi, \\ w = \frac{2P}{m J_0} = \chi_d - b, \qquad w' = |u|^2, \quad b' = -|u|^2. \end{gathered} \tag{5}$$

Initially $b = \kappa$. In particular

$$\frac{m_f}{m_0} = \frac{\chi}{\chi + J}, \qquad J \le \kappa. \tag{6}$$

This inverse-mass state makes resource propagation additive, including across coasts and unpowered encounters.

1.2 Ideal and bounded engines

The ideal problem permits any square-integrable planar u. The full-power/coast engine instead admits

$$\begin{gathered}\mathcal{U}(b) = \{0\} \cup \{u : w/\nu_+ \le |u| \le w/\nu_-\}, \\ \nu_\pm = V_{e,\pm}/V_0.\end{gathered} \tag{7}$$

At $b = 0$ only coast remains. A powered acceleration approaching zero requires unbounded exhaust speed, so an ideal solution need not obey a finite ceiling even when most of its burn does.

The ideal circular-endpoint action depends on (ρ, Θ, h), where the departure radius is the length scale and ρ is the arrival radius ratio. Budget κ determines affordability afterward. The bounded action additionally depends on χ, ν_-, ν_+ and the available reserve. Fixing the band ratio removes one coordinate but does not remove χ. If $\mu, V_{e,+}$ and initial specific useful power $\beta_{\text{jet}} = P/m_0$ are all fixed across a vehicle family, the restricted relation

$$\chi = \frac{2\beta_{\text{jet}}\mu}{V_{e,+}^5}\nu_+^5 \tag{8}$$

removes one more coordinate. Changing fuel load at fixed hardware power does not preserve that family assumption.

1.3 A free-space benchmark

For $g = 0$, fixed initial and final states and duration h, set $d_r = r_f - r_0 - hv_0$, $d_v = v_f - v_0$. The minimum effort is

$$J_{\text{fs}} = \frac{12|d_r|^2}{h^3} - \frac{12d_r \cdot d_v}{h^2} + \frac{4|d_v|^2}{h}. \tag{9}$$

To see this, write the endpoint constraints as the two moments of u, whose Gramian is

$$W_0 = \begin{pmatrix} h^3 I/3 & h^2 I/2 \\ h^2 I/2 & hI \end{pmatrix}. \tag{10}$$

The minimum-norm control is affine in time, and its squared norm is $(d_r, d_v)^T W_0^{-1}(d_r, d_v)$, giving (9). For a dimensional rest-to-rest displacement L,

$$a(t) = \frac{6L}{T^2}\left(1 - \frac{2t}{T}\right), \qquad j = \frac{12L^2}{T^3}. \tag{11}$$

Consequently $T_{\min} = [6L^2/(P(m_d^{-1} - m_0^{-1}))]^{1/3}$ in this ideal free-space problem. Gravity changes the endpoint moments, so (9) is not in general a lower bound on Kepler effort.

1.4 Provenance, scope and conventions

The starting points are the author's constant-power Kepler transfer and rocket pursuit–evasion articles [12, 11]. Inherited results include inverse-mass accounting, ideal free-space transfers, the Kepler similarity reduction, interception and deadline-fuel games, and the inertial gravity-compensation certificate. This article develops the moving-endpoint derivatives F_h, F_d, the nonconvexity example, the bounded resource adjoint on mass-dependent exhaust limits, the radial-measure specialization, flyby and SOI junctions, the circular-spiral model, and the CW rendezvous game. These are extensions or specializations of classical optimal-control and game theory, not replacements for it [15, 10, 8]. The planar restriction is deliberate. The chase article [11] also treats $d = 3$; the inertial free-space results below hold in either dimension.

Numerical data taken from the earlier articles come from repository commit `bdc9698212c92cd0815d9251c7d777d47cefd706`. All proofs needed for the stated analytical results are included here. Numerical scripts and input data are provided as supplementary material; a numerical observation is distinguished from a proof.

Convention	Translation
Kepler time Hamiltonian	$H_T = \lambda \cdot f - 1$, maximized; mass is a state
Ideal thrust gain	$C_m = \lambda_m m^2 > 0$, $k_T = P/C_m$
Physical effort costates	$p_{r,\text{phys}} = 2k_T\lambda_r$, $p_{v,\text{phys}} = 2k_T\lambda_v$
Normalized costates	$p_r = (R_0/J_0)p_{r,\text{phys}}$, $p_v = (V_0/J_0)p_{v,\text{phys}}$
Chase resource	$b_{\text{chase}} = J_0 b$ in m^2/s^3
Kepler capability	$\kappa = \chi_d - \chi$
Mass ratio	$\eta_m = \chi_d/\chi = m_0/m_d$ (Kepler's η)
Exhaust limits	$\nu_\pm = V_{e,\pm}/V_0$
Minimizing DP convention	$p = -\nabla_s U$ along a differentiable optimal cost-to-go; also convert objective units

The costate gain conversion is for an ideal interior arc: there k_T is constant and $a = k_T\lambda_v = p_{v,\text{phys}}/2$. Applying the same constant conversion across a capped boundary is invalid. Multipliers for mass or resource endpoints must be transformed with their state coordinate.

Symbol	Meaning and separation
β_{jet}	P/m_0, W/kg; distinct from hardware coefficient β_{sys}, kg/W
Dg	Jacobian of gravity; D in the variational matrix is $BB^T/2$
$D(s)$	Flyby turn bound; D in the interception section is separation beyond capture radius
D in the spiral	Progress $\lvert q_i - q_f\rvert$ in Section 5; distinct from the turn bound and matrix block
C	Resource coefficient $p_b + \ell_2$; $C(h)$ is a rendezvous effort; C denotes coast
ω	Target mean motion in timing; local comparison frequency in verification, defined there; primer slope below is n
$L, \mathcal{L}$	Physical length (or a labeled lower bound), and the angular costate integral, respectively
η_m, η	Mass ratio and residual-error coefficient in the global-gap test, respectively

1.5 Power–mass closure

For design studies, let $m_d = m_{\text{pay}} + \beta_{\text{sys}} P$ and fix $\mathcal{R} = m_{\text{prop}}/m_d$. A useful hardware model is

$$\beta_{\text{sys}} = \frac{1}{(1-\varphi)\alpha_{\text{eng}}} + \frac{\varphi}{(1-\varphi)\rho_{\text{rad}}},$$

where φ is the rejected fraction of input power, α_{eng} the engine specific input power and ρ_{rad} the radiator specific rejected power. Both specific powers are in W/kg. Substituting $m_0 = (1+\mathcal{R})m_d$ into the ideal rest-to-rest formula gives

$$T^3 = \frac{6L^2(1+\mathcal{R})}{\mathcal{R}} \left(\frac{m_{\text{pay}}}{P} + \beta_{\text{sys}} \right). \tag{12}$$

Thus $T_\infty = [6L^2(1+\mathcal{R})\beta_{\text{sys}}/\mathcal{R}]^{1/3}$, $P_{\text{knee}} = m_{\text{pay}}/\beta_{\text{sys}}$, and $T = T_\infty(1 + P_{\text{knee}}/P)^{1/3}$. This inherited hardware closure [12] is an ideal free-space design benchmark, not a finite-band orbital prediction. It complements the fixed-hardware mass identity without identifying W/kg with kg/W.

2 Endpoint action and orbital timing

2.1 One Hamiltonian convention

Use the maximizing Hamiltonian with no factor 1/2 in effort:

$$\begin{gathered} H = p_r \cdot v + p_v \cdot (g+u) - |u|^2, \\ u = p_v/2, \qquad H^* = p_r \cdot v + p_v \cdot g + |p_v|^2/4, \\ p_r' = -Dg(r)^T p_v, \qquad p_v' = -p_r. \end{gathered} \tag{13}$$

On a smooth autonomous ideal phase, H^* and $\mathcal{L} = p_r \cdot \mathsf{J}r + p_v \cdot \mathsf{J}v$ are constant, where $\mathsf{J}(x,y) = (-y,x)$. These are costate integrals, distinct from mechanical energy and angular momentum. The velocity costate is the primer vector, with the effort normalization $u = p_v/2$ used here [10].

For a regular stationary action branch $S(s_0, s_f, t_0, t_f)$, with $s = (r, v)$, endpoint variation gives

$$dS = p_f \cdot ds_f - p_0 \cdot ds_0 - H_f dt_f + H_0 dt_0. \tag{14}$$

This is the central sensitivity relation [3]. It is a value derivative only where that branch is minimizing and the value is differentiable.

For circular states $s_f = (\rho e_r(\Theta), \rho^{-1/2} e_\theta(\Theta))$, the intrinsic action $\mathcal{E}(\rho, \Theta, h)$ satisfies

$$\begin{gathered} \mathcal{E}_h = -H, \qquad \mathcal{E}_\Theta = \mathcal{L}, \\ \mathcal{E}_\rho = p_{r,f} \cdot e_r - \tfrac{1}{2}\rho^{-3/2} p_{v,f} \cdot e_\theta. \end{gathered} \tag{15}$$

An angle variation rotates both position and velocity. Cartesian endpoints are periodic in Θ; the winding label must be tracked separately.

2.2 Moving-endpoint derivatives

Let d, a be departure and arrival epochs, $h = a - d$, and let ϕ be the target's phase at epoch zero. After rotating the departure state to angle zero,

$$\Theta = \phi + \omega a - d + 2\pi N, \qquad \omega = \rho^{-3/2}. \tag{16}$$

For arbitrary prescribed endpoint ephemerides,

$$\begin{aligned} \mathcal{F}_d\big|_a &= H_0 - p_0 \cdot \dot{s}_D(d), \\ \mathcal{F}_a\big|_d &= p_f \cdot \dot{s}_A(a) - H_f. \end{aligned} \tag{17}$$

If the endpoint ephemerides follow the same unforced dynamics as the adjacent ideal phase, $H = p \cdot f_{\text{ballistic}} + |u|^2$. Thus, for $F(d,h) = \mathcal{F}(d, d+h)$,

$$\boxed{F_h = -|u_f|^2, \qquad F_d = |u_0|^2 - |u_f|^2.} \tag{18}$$

The geometrically fixed-endpoint derivative is $-H$; the moving-arrival derivative is $-|u_f|^2$. Confusing them changes the time-search direction. For circular endpoints $F_d = (\omega - 1)\mathcal{L}$. At an interior optimal departure date with fixed duration, $|u_0| = |u_f|$.

Second derivatives require variational data. Writing $A = \left(\begin{smallmatrix} 0 & I \\ Dg & 0 \end{smallmatrix}\right)$, $B = (0, I)^T$, $D = BB^T/2$, and $C_s = \operatorname{diag}(K, 0)$ with $K = D_r^2(p_v \cdot g)$, the state–costate variation obeys

$$\begin{pmatrix} \delta s \\ \delta p \end{pmatrix}' = \begin{pmatrix} A & D \\ -C_s & -A^T \end{pmatrix} \begin{pmatrix} \delta s \\ \delta p \end{pmatrix}. \tag{19}$$

Let its transition blocks be $M_{ss}, M_{sp}, M_{ps}, M_{pp}$. For fixed initial state, solve $M_{sp}\delta p_0 = \delta s_f - f_f \delta h$. For a moving ballistic arrival, the right side per unit duration is simply $-Bu_f$. Therefore

$$F_{hh} = -\tfrac{1}{2} p_{v,f} \cdot \left[M_{pp}(-M_{sp}^{-1} B u_f) + \dot{p}_f \right]_v. \tag{20}$$

Indeed the total endpoint-costate derivative is $dp_f/dh = M_{pp}(dp_0/dh) + \dot{p}_f$. Differentiating $F_h = -p_{v,f} \cdot p_{v,f}/4$ gives the displayed expression. Use a linear solve and monitor conditioning. A singular block indicates a failing branch chart, not a value-gradient interpolation problem that can be repaired by smoothing.

2.3 From effort to time and fuel

Appending an admissible coast with the destination proves that the global moving-endpoint effort value $F^*(d,h)$ is nonincreasing in h. The argument needs position and velocity matching and an admissible destination orbit. It survives engine limits and safety constraints when those conditions hold. The earliest feasible duration is

$$h^*(\kappa) = \inf\{h > 0 : F^*(d,h) \le \kappa\}. \tag{21}$$

On a regular ideal branch with $F_h \neq 0$,

$$\frac{dh^*}{d\kappa} = -\frac{1}{|u_f|^2}, \qquad h_{\text{new}} = h + \frac{F(d,h)-\kappa}{|u_f|^2}. \tag{22}$$

At the value level this recovers the original minimum-time problem exactly: a trajectory exhausts its permitted fuel precisely when its effort is κ, and $F^* \le \kappa$ is exactly existence of an affordable transfer when the fixed-time infimum is attained. At an unattained nonconvex boundary, use existence of a control with cost at most the budget as the feasibility predicate, rather than an infimum alone. For the original *fixed final-angle* targeting problem, use $F(h) = \mathcal{E}(\rho, \Theta_{\text{fixed}}, h)$ in the same feasibility infimum, with $F_h = -H$; monotonicity and the displayed moving-target Newton denominator are not thereby implied. For a prescribed initial planet phase, use $\Theta(h) = \phi + \omega h$, and the ballistic-endpoint formulas above apply. At a specified duration these are the same fixed-time problem when their endpoint states agree.

For an ideal normal Kepler time extremal ending at dry mass, the constant rescaling in the convention table gives the fixed-time effort PMP at $h = t_f/T_0$, including the same endpoint conditions. An extremal is thereby an effort extremal, not necessarily a minimum. For a global time minimizer on a regular continuous frontier, any strictly cheaper fixed-time transfer would retain budget slack under a small earlier-arrival endpoint correction, contradicting earliest arrival. Hence $F^*(d,h^*) = \kappa$ there and the original solution is also a fixed-time effort minimizer. This last implication requires that endpoint regularity; the infimum formula itself does not.

Safeguard the step within a feasible/infeasible bracket. A stationary branch can give an affordable candidate without locating the globally earliest arrival.

In physical units, at fixed m_0, P,

$$\frac{dm_{\text{prop}}}{dT} = -\frac{m_f^2}{2P}|a_f|^2. \tag{23}$$

For prescribed final mass and fixed P, the required launch mass is $m_0 = m_d/(1 - m_d j/(2P))$, provided $j < 2P/m_d$. For a redesigned family with $P/m_0 = \beta_{\text{jet}}$ instead, $m_0 = m_d(1 + j/(2\beta_{\text{jet}}))$. These load-selection problems have different derivatives and must use different vehicle parameter updates.

At a freely timed velocity-matched intermediate stop, adjacent ideal legs with equal power satisfy $dJ_{\text{total}}/dt_j = |u_{j+1,0}|^2 - |u_{j,f}|^2$ in common units. Different constant powers require the physical weights $1/(2P_j)$. For bounded engines, mass couples the leg actions; the augmented endpoint relation (14) is required.

2.4 Monotone does not mean convex

The global moving-endpoint value need not be convex, even near a circular coast and without switching between branches. Consider an arrival radius $1+\epsilon$ and initial phase ϵ. Linearizing around the unit coast gives

$$\begin{aligned} F_\epsilon^*(h) &= \epsilon^2 c(h) + O(\epsilon^3), \\ c(h) &= [\Phi(h)\xi]^T W(h)^{-1}[\Phi(h)\xi], \end{aligned} \tag{24}$$

where $\xi = (1,1,0,-3/2)^T$ and Φ, W are the CW transition matrix and Gramian in Section 7. The coefficients are

h	$c(h)$
10	0.0829955305949018
10.5	0.0816445096953235
11	0.0801408830697246

(25)

and the midpoint exceeds the endpoint average by $7.6302863010368\,10^{-5} > 0$. Appendix E supplies the interval calculation and its remainder bounds.

To justify the global expansion, CW controllability makes the endpoint derivative onto at zero control; the reduced cost Hessian there is $2I$. The nearby stationary branch has effort $O(\epsilon^2)$. Any cheaper competitor has control norm $O(\epsilon)$ and, by continuous dependence on a finite horizon, stays near the same nonsingular coast. On that small control ball the Lagrangian Hessian is $2I+O(\epsilon)$, hence positive. This establishes the expansion of the global value at the three horizons for sufficiently small positive ϵ, and the strict Jensen violation follows. No numerical upper threshold on ϵ is asserted. Time search should exploit monotone feasibility, not assume convexity.

3 Engine limits and the solar floor

3.1 The bounded-engine selector

For normal running cost $\ell_0 + \ell_2|u|^2$, set $C = p_b + \ell_2$. Fixed-time effort uses $(\ell_0, \ell_2) = (0,1)$; minimum time uses $(1,0)$. The Hamiltonian is

$$H = p_r \cdot v + p_v \cdot g - \ell_0 + p_v \cdot u - C|u|^2. \tag{26}$$

Let $s = |p_v|$, $a_- = w/\nu_+$ and $a_+ = w/\nu_-$. For $C > 0$,

$$\widehat{a} = \text{clip}(s/(2C), a_-, a_+), \qquad \mathcal{G} = \max(0, s\widehat{a} - C\widehat{a}^2). \tag{27}$$

Burn in direction p_v/s when the powered gain is positive and coast when it is negative. At a tie both maximize. For $C \le 0$, compare coast with a_+. If $s = 0$ and a burn wins, its direction is not determined by this pointwise condition.

For $C > 0$ the four regular magnitude regions are

$$a^* = \begin{cases} 0 & s < Ca_-, \\ a_- & Ca_- < s < 2Ca_-, \\ s/(2C) & 2Ca_- < s < 2Ca_+, \\ a_+ & s > 2Ca_+. \end{cases} \tag{28}$$

The lowest powered thrust uses maximum exhaust; the highest uses minimum exhaust. State and costate evolution determines the order of these modes along a transfer.

The kinematic adjoints retain (13), but the resource adjoint changes on a fixed-exhaust boundary:

$$\begin{gathered} C' = \frac{s - 2Ca}{\nu} \quad (a = w/\nu), \\ C' = 0 \quad \text{on interior burns and coasts.} \end{gathered} \tag{29}$$

Indeed $\partial a/\partial b = -1/\nu$, so differentiation of the optimized Hamiltonian gives (29). Ordinary switches inside $b > 0$ have continuous states and costates. Dry-mass exhaustion is a state-constraint event and requires its additional multiplier. For free inactive terminal reserve in an isolated effort problem, $p_b(h) = 0$; an embedded leg receives its resource covector from the next leg.

Autonomy and rotational symmetry preserve H^* and $\mathcal{L}$ on the regular phase. The product $\lambda_m m^2$ does not stay constant on capped arcs, because the acceleration bounds depend on mass. Appendix A gives the one-dimensional propagation. There is no derived universal finite switching sequence for the Kepler problem.

With ordinary ballistic endpoint ephemerides, the bounded counterparts of (18) are

$$F_h = -\mathcal{G}_f, \qquad F_d = \mathcal{G}_0 - \mathcal{G}_f, \tag{30}$$

when terminal resource variations vanish or are included through their transversality conditions. The ideal case has $p_b = 0$ and $\mathcal{G} = |u|^2$.

3.2 Auditing an ideal trajectory

Given an endpoint-exact ideal path, write $I(t) = \int_0^t |u|^2 ds$ and $w_I = \chi + I$. Its actual full-power exhaust on a burn is $\nu_e = w_I/|u|$. It is band-feasible precisely when $I(h) \le \kappa$ and

$$w_I/\nu_+ \le |u| \le w_I/\nu_- \tag{31}$$

almost everywhere on powered intervals; exact coasts pass separately. Certified interval bounds can establish (31) between samples. Clipping a violating control changes its endpoint. If a globally ideal-minimal path passes this audit, it is also globally minimal in the restricted problem at that time.

Continuous throttling is a different engine model. For duty fraction δ, $a = \delta w/\nu$ and $w' = a^2/\delta$. The fuel-minimal realization of a prescribed acceleration history uses

$$\nu = \min(\nu_+, w/a), \qquad w' = \max(a^2, wa/\nu_+). \tag{32}$$

It also requires $a \le w/\nu_-$ and $w(h) \le \chi_d$. This scalar equation preserves the prescribed spatial path exactly under continuous throttling. An on/off approximation must preserve both mean acceleration and mass flow: a duty cycle has mean effort rate δa_{burn}^2, not $(\delta a_{\text{burn}})^2$. Exact endpoint correction for finite pulses is an additional problem.

3.3 A hard radial constraint

Set $R_c = r_{\min}/R_0$ and impose $c = (R_c^2 - |r|^2)/2 \le 0$. For the solar model $r_{\min} = 0.1$ AU; $R_c = 0.1$ only when $R_0 = 1$ AU. With radial and tangential speeds y, z,

$$\begin{gathered} R' = y, \qquad y' = z^2/R - 1/R^2 + u_r, \\ \theta' = z/R, \qquad z' = u_\theta - yz/R. \end{gathered} \tag{33}$$

The constraint has relative order two: $c' = -r \cdot v$ and $c'' = -|v|^2 + 1/R - r \cdot u$. At a boundary interval,

$$\begin{gathered} R = R_c, \quad y = 0, \quad z' = u_\theta, \\ u_r = A(z) := R_c^{-2} - z^2/R_c. \end{gathered} \tag{34}$$

A circular coast with $z = \pm R_c^{-1/2}$ is a zero-effort globally minimal boundary arc for its own endpoints. Thus boundary intervals must be permitted.

Using the original measure-costate convention, a nonnegative measure dM is supported on the contact set $[6, 1]$:

$$dp_r = -Dg^T p_v \, dt - r \, dM, \qquad dp_v = -p_r \, dt. \tag{35}$$

At an atom $\gamma \ge 0$, $p_r^+ = p_r^- - \gamma r$ and $p_v^+ = p_v^-$. The ideal thrust is continuous, while its derivative can jump. Tangency makes the Hamiltonian continuous; rotational symmetry preserves $\mathcal{L}$. Higher-order transformed adjoints use a different representation and should not be mixed with these jumps.

For the ideal boundary problem, substituting (34) gives $J_B = \int (A(z)^2 + z'^2) dt$ and

$$z'' = -\frac{\mathcal{L}}{2R_c} - \frac{2A(z)z}{R_c}. \tag{36}$$

The original radial adjoint fixes its measure density:

$$\mu_M = \frac{-4A/R_c^3 - 6z'^2/R_c - 8zz''/R_c - 2Az^2/R_c^2}{R_c} \ge 0. \tag{37}$$

For completeness, on the circle $p_v = 2Ae_r + 2z'e_\theta$ and $e_r' = (z/R_c)e_\theta$ give $p_r = -p_v' = 6zz'e_r/R_c - 2(z'' + Az/R_c)e_\theta$. The radial component of $p_r' = -Dg^T p_v - R_c e_r \mu_M$, with $Dg\, e_r = 2e_r/R_c^3$, gives (37) term by term. This sign test and the entry/exit conditions are part of the solution. For constant z, a subcircular boundary burn has $\mu_M < 0$ and fails the normal necessary conditions.

With engine bounds the boundary is feasible only if $|A| \le a_+$ and

$$\sqrt{\max(0, a_-^2 - A^2)} \le |u_\theta| \le \sqrt{a_+^2 - A^2}. \tag{38}$$

Coast is separately possible when $A = 0$. Add the radial measure to the bounded-engine adjoints and retain (29).

A smooth radial penalty preserves time and rotation symmetry but changes the objective and its sensitivities. It does not in general enforce the hard floor. The barrier $\epsilon/(R - R_c)^2$ on $R > R_c$ excludes contact at finite penalized cost and can initialize continuation, but cannot itself represent a boundary interval. For a regular constrained minimum with fixed endpoints,

$$\frac{\partial J^*}{\partial R_c} = R_c M([0, h]) \ge 0. \tag{39}$$

The physical effort must be reported separately from any penalty cost.

For runtime safety, a cubic Bezier position interpolant p with controls B_i has squared-radius Bernstein coefficients

$$c_k = \sum_{i+j=k} \frac{\binom{3}{i}\binom{3}{j}}{\binom{6}{k}} B_i \cdot B_j. \tag{40}$$

If a validated trajectory error is $|r - p| \le E$, then $\min_k c_k \ge (R_c + E)^2$ is a sufficient segment pass. Subdivide unresolved segments. The error enclosure must include trajectory and arithmetic error, not just interpolation of sampled data.

4 Joining transfers through planetary encounters

4.1 An unpowered point event

In the zero-duration patched-conic model, let the planet state at encounter epoch τ be (R_p, V_p) with acceleration A_p. Use common heliocentric units, incoming excess velocity $w_- = v^- - V_p$, speed $s = |w_-| > 0$, and rotation $Q(\alpha)$. The event map is

$$r^- = r^+ = R_p, \quad v^+ = V_p + Q(\alpha) w_-, \quad b^+ = b^-. \tag{41}$$

The position is a mathematical encounter node, not a physical collision. With normalized planetary parameter $\bar{\mu}_p$ and minimum planetocentric periapsis $q_{\min}$,

$$|\alpha| \le D(s) = 2 \arcsin(1/e), \qquad e = 1 + \frac{q_{\min} s^2}{\bar{\mu}_p}. \tag{42}$$

This preserves excess speed and consumes no propellant. The heliocentric energy change is $\Delta E_\odot = V_p \cdot (Q w_- - w_-)$, and

$$|\Delta v| \le \frac{2s}{1 + q_{\min} s^2/\bar{\mu}_p} \le \sqrt{\bar{\mu}_p / q_{\min}}. \tag{43}$$

Zero excess speed is not a hyperbolic flyby. Zero turn is an infinite-periapsis limit of this event set; a finite-SOI model treats the geometry explicitly.

A circular-rendezvous atlas fixes the ship velocity to the planet's, so it has $w_- = 0$ at that endpoint and cannot supply general flyby legs. General endpoint actions are needed. Once they exist, one encounter can be parameterized by $(w_{-,x}, w_{-,y}, \alpha, \tau)$. For the sum F of incoming and outgoing actions, let superscripts $-, +$ denote their adjacent endpoint covectors and put $w_+ = Q w_-$. Equation (14) gives

$$\begin{aligned} F_{w_-} &= p_v^- - Q^T p_v^+, \\ F_\alpha &= -p_v^+ \cdot \mathsf{J} w_+, \\ F_\tau &= H^+ - H^- + (p_r^- - p_r^+) \cdot V_p \\ &\qquad + (p_v^- - p_v^+) \cdot A_p. \end{aligned} \tag{44}$$

Here the planet-relative w_- is held fixed when varying epoch; both endpoint velocities acquire $A_p\, d\tau$. Specifically, the incoming endpoint differential is $p_r^- \cdot V_p\, d\tau + p_v^- \cdot A_p\, d\tau - H^- d\tau$; the outgoing initial differential has the opposite covector signs and $+H^+ d\tau$. Their sum is the stated F_τ.

Let $\eta_\pm \ge 0$ multiply $\pm\alpha - D(w_-) \le 0$ and $d_D = \nabla D = -4(q_{\min}/\bar{\mu}_p) w_- / (e\sqrt{e^2 - 1})$. Interior-event stationarity requires

$$\begin{aligned} p_v^- &= Q^T p_v^+ + (\eta_+ + \eta_-) d_D, \\ p_v^+ \cdot \mathsf{J} w_+ &= \eta_+ - \eta_-, \qquad F_\tau = 0, \end{aligned} \tag{45}$$

together with complementary slackness. An interior turn need not be maximum turn. At a prescribed periapsis and turn side $\sigma = \pm 1$, the velocity Jacobian is

$$M = Q + (\mathsf{J} w_+)(\sigma d_D)^T, \qquad p_v^- = M^T p_v^+. \tag{46}$$

The extra term differentiates the speed-dependent turn; $\det M = 1$. Resource costates match if the same resource passes unchanged. The position guard supplies an independent position-covector multiplier. The matched Hamiltonian combination is $H - p_r \cdot V_p - p_v \cdot A_p$, rather than H alone.

4.2 A finite powered encounter

For an encounter with duration, use a planet-centered nonrotating patch, length scale L and local units $T_p = \sqrt{L^3/\mu_p}$, $V_p^* = \sqrt{\mu_p/L}$, $J_p = (V_p^*)^2/T_p$. The controlled equations again have form (4), with x the relative position and

$$\varepsilon \le |x| \le \sigma_S, \quad \varepsilon = r_{p,\min}/L, \quad \sigma_S = R_S/L. \tag{47}$$

Choose inward entry and the first outward exit. With $L = R_S$, $\sigma_S = 1$. Distinct loop branches require explicit labels. The engine selector and radial-floor conditions are those of Section 3 in the new units.

At an endpoint with positive planetary energy $E = |v|^2/2 - 1/|x|$, put $s = \sqrt{2E}$, $j = x \times v$ and

$$e_v = (|v|^2 - 1/|x|)x - (x \cdot v)v, \quad e = |e_v|, \quad n = e_v/e. \tag{48}$$

The hypothetical ballistic tails have excess-velocity labels

$$w_{\infty,-} = \frac{s}{e}(n + js\mathsf{J}n), \qquad w_{\infty,+} = \frac{s}{e}(-n + js\mathsf{J}n). \tag{49}$$

Use the incoming expression at entry and outgoing expression at exit. Thrust can change their magnitudes. The unpowered equal-speed and maximum-turn constraints must not be imposed on this controlled phase. An $E \le 0$ endpoint has no hyperbolic excess label, although finite-SOI crossing can still occur.

The exact work and angular-momentum identities are

$$E' = v \cdot u, \qquad j' = x \times u. \tag{50}$$

For positive-energy labels, $(s_+^2 - s_-^2)/2 = \int v \cdot u\, dt$. High periapsis speed increases instantaneous work per thrust increment when thrust is suitably aligned. The mission optimum must also satisfy exit geometry, timing and resource conditions. It does not follow by maximizing this work alone. An impulse is not a constant-power burn: compressing a nonzero Δv into duration δ costs at least $|\Delta v|^2/\delta$. Likewise infinity-to-infinity hyperbolic flight

has infinite duration; the optimized mission segment has finite SOI endpoints.

On a connected unconstrained ideal phase, $p_v'' = Dg(x)^T p_v$. A positive-length coast forces $p_v = p_v' = 0$ there, hence everywhere by uniqueness. A nontrivial powered ideal extremal therefore has no positive-length coast interval. The finite-band selector admits coasts without $p_v = 0$ and has no such restriction.

4.3 The moving SOI

Return to one common physical or heliocentric unit system. For ship coordinates (X, V) put $x = X - R_p(t)$ and $v_{\rm rel} = V - V_p(t)$. The consistent global patch fields are

$$G_\odot = -\mu_\odot X/|X|^3, \qquad G_P = A_p - \mu_p x/|x|^3. \quad (51)$$

Subtracting the origin acceleration gives the desired pure planetary relative dynamics. Solar tides inside the patch are omitted.

The SOI guard is $c_S = (|x|^2 - R_S^2)/2 = 0$. At a free transverse crossing with continuous state and mass, endpoint variation gives

$$\begin{gathered} p_X^+ = p_X^- + \zeta x, \quad p_V^+ = p_V^-, \quad p_b^+ = p_b^-, \\ H^+ - H^- = \zeta x \cdot V_p. \end{gathered} \quad (52)$$

These are hybrid junction conditions [13]. The equality multiplier ζ has no sign restriction; it is different from a radial exclusion multiplier. With unchanged engine and a unique regular selector,

$$\zeta = -\frac{p_V \cdot (G^+ - G^-)}{x \cdot v_{\rm rel}}. \quad (53)$$

To derive this, substitute $\Delta p_X = \zeta x$ into the Hamiltonian jump: $\Delta H = \zeta x \cdot V + p_V \cdot \Delta G$. Equating it to $\zeta x \cdot V_p$ and dividing by the nonzero transverse speed $x \cdot (V - V_p)$ yields (53). Grazing invalidates this chart. Coincident constraint events require their additional multipliers.

A useful optimization chart is (θ, v_r, v_t, τ): $X = R_p + R_S e_r(\theta)$, $V = V_p + v_r e_r + v_t e_\theta$. With $d_r = p_X^- - p_X^+$, $d_v = p_V^- - p_V^+$,

$$\begin{aligned} F_\theta &= R_S d_r \cdot e_\theta + d_v \cdot (v_r e_\theta - v_t e_r), \\ F_{v_r} &= d_v \cdot e_r, \qquad F_{v_t} = d_v \cdot e_\theta, \\ F_\tau &= H^+ - H^- + d_r \cdot V_p + d_v \cdot A_p. \end{aligned} \quad (54)$$

At a regular optimized crossing these derivatives vanish. The position jump in (52) is normal to the surface, as required by the angle variation.

Translation changes the Hamiltonian to

$$H_{\rm rel} = H_{\rm global} - p_X \cdot V_p - p_V \cdot A_p, \quad (55)$$

which is continuous at the interface. For a given continuous control the state variation also has saltation matrix

$$\begin{gathered} S = I + \frac{(f^+ - f^-)n^T}{n^T f^- + c_{S,t}}, \\ n = (x, 0, 0), \qquad f^+ - f^- = (0, \Delta G, 0). \end{gathered} \quad (56)$$

It reproduces the covector jump through $p^- = S^T p^+$. Full optimal-control shooting derivatives additionally need the selector and adjoint variations.

4.4 Units and mass across a route

Let $\lambda = L/R_0$ and $\gamma_\mu = \mu_p/\mu_\odot$. The local-to-Sun ratios are

$$c_t = \frac{\lambda^{3/2}}{\sqrt{\gamma_\mu}}, \quad c_v = \sqrt{\gamma_\mu/\lambda}, \quad c_J = \frac{\gamma_\mu^{3/2}}{\lambda^{5/2}}. \quad (57)$$

For actions normalized by their respective effort scales,

$$p_X^{(0)} = \frac{c_J}{\lambda} p_x^{(p)}, \quad p_V^{(0)} = \frac{c_J}{c_v} p_v^{(p)}, \quad H^{(0)} = \frac{c_J}{c_t} H^{(p)}. \quad (58)$$

The resource covector is unchanged when resource and action use the same effort scale. Apply any frame rotation as well. This follows by preserving the physical endpoint differential, not by matching raw normalized covectors.

For equal power on every leg, total physical effort and time are sums $\sum_i J_{0,i} J_i$ and $\sum_i T_{0,i} h_i$. For unequal constant powers without mass resets,

$$\frac{1}{m_f} - \frac{1}{m_0} = \sum_i \frac{J_{0,i} J_i}{2P_i}. \quad (59)$$

Each leg starts at the preceding leg's actual final mass. Resource states cannot be reset to their original launch values. A moon patch uses its prescribed origin acceleration and the actual parent field outside; do not add the origin acceleration twice.

A prescribed stop, a point flyby and an SOI crossing have different interface conditions. Once all interior variables are stationary, the envelope theorem leaves only the outer endpoint terms in a route timing derivative. It does not make each embedded phase satisfy $H = 0$.

5 Planetary escape and capture

For a station on a circular parking orbit, take its radius as the local length scale. The station state is $x_s(t) = e_r(\phi_s + t)$, $v_s(t) = e_\theta(\phi_s + t)$. Escape starts at that state and ends at the first outward SOI crossing. Capture starts at an inward crossing and must match both station position and velocity at the actual arrival epoch. Matching the orbital radius alone is insufficient.

5.1 A solvable circular approximation

For a slowly evolving circular orbit let $q = r^{-1/2}$. Tangential acceleration gives the leading-order equations

$$q' = -a_t, \qquad \theta' = q^3, \qquad r' = 2r^{3/2} a_t. \quad (60)$$

The validity diagnostic is $\eta_{\rm ad} = |a_t| r^2 \ll 1$; the radial-to-circular-speed ratio is $2\eta_{\rm ad}$. This approximation assigns radial drift during a burn and cannot be attached literally to an exactly circular initial state with zero radial speed.

Assume monotone radius and free phase, put $D = |q_i - q_f|$ and $a = |a_t|$. Then $\int a\, dt = D$. The ideal minimum effort is D^2/h, attained by constant $a = D/h$. For the finite band, $w' = a^2$ and the final inverse mass

is $W = \chi + J$. On powered intervals use $z = 1/a$ and w as the independent variable:

$$\int_\chi^W z\,dw = D, \qquad T_b = \int_\chi^W z^2\,dw, \\ \nu_-/w \le z \le \nu_+/w. \tag{61}$$

Coasts consume neither progress nor mass in this model.

Proposition 1 (Finite-band circular profile)**.** *For fixed W and feasible monotone progress D, the unique minimum-burn-time profile almost everywhere is*

$$z_*(w) = \mathrm{clip}(k, \nu_-/w, \nu_+/w), \qquad \int_\chi^W z_*\,dw = D, \tag{62}$$

where feasibility is $\nu_- \log(W/\chi) \le D \le \nu_+ \log(W/\chi)$.

Proof. For a fixed scalar k, pointwise projection minimizes $(z-k)^2$ on the interval. Integrating and comparing any other feasible z, the equality of $\int z\,dw$ cancels the linear term, leaving $\int z_*^2 dw \le \int z^2 dw$. Strict convexity gives uniqueness. The integral of the clipped profile is continuous and monotone in k. □

As w increases the profile uses, when present, minimum exhaust, constant acceleration and maximum exhaust. This ordering is specific to the reduced model. Define

$$w_1 = \mathrm{clip}(\nu_-/k, \chi, W), \\ w_2 = \mathrm{clip}(\nu_+/k, w_1, W). \tag{63}$$

The scalar root uses the elementary formulas

$$D = \nu_- \log(w_1/\chi) + k(w_2 - w_1) + \nu_+ \log(W/w_2), \\ T_b = \nu_-^2(\chi^{-1} - w_1^{-1}) + k^2(w_2 - w_1) \\ \quad + \nu_+^2(w_2^{-1} - W^{-1}). \tag{64}$$

For constant exhaust ν,

$$J_\nu = \chi(e^{D/\nu} - 1), \qquad T_\nu = \frac{\nu^2}{\chi}(1 - e^{-D/\nu}). \tag{65}$$

Set $T_{\mathrm{fast}} = T_{\nu_-}$ and $T_{\mathrm{slow}} = T_{\nu_+}$. For $D > 0$, duration below T_{fast} is infeasible in the reduced model. Between these times there is a unique effort giving $T_b = h$; above T_{slow} the optimum is a maximum-exhaust burn plus coast, with the positive floor J_{ν_+}. At fixed D, the interior envelope derivative of burn time with respect to W is $z_*(W)^2 - 2kz_*(W) < 0$, establishing the outer monotone effort root. For $D = 0$, zero effort is sufficient when phase is free.

Station phase adds a constraint. For constant acceleration the accumulated angle is

$$\Delta\theta = \frac{h}{4}(q_i^3 + q_i^2 q_f + q_i q_f^2 + q_f^3). \tag{66}$$

It is not a free output. On the maximum-exhaust plateau, let $c = h - T_{\mathrm{slow}}$ and let Θ_b be the burn angle. Placing coast at the initial and final radii gives exactly the range

$$\Theta_b + c\min(q_i^3, q_f^3) \le \Delta\theta \le \Theta_b + c\max(q_i^3, q_f^3) \tag{67}$$

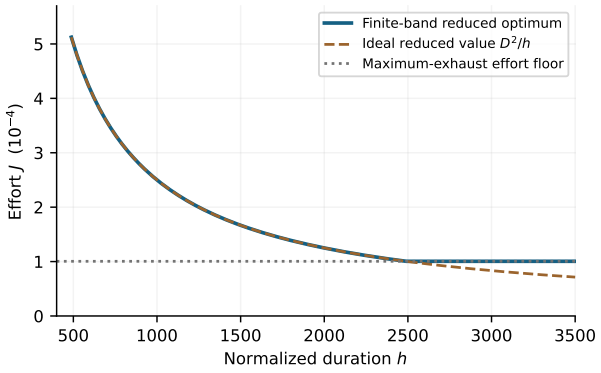


Figure 1: Minimum effort in the phase-free monotone circular model: $D = 0.5$, $\chi = 0.01$, $\nu_- = 10$, $\nu_+ = 50$. The finite ceiling produces a long-duration fuel floor.

at that minimum fuel, within the monotone approximation. Outside this range phase costs additional effort or makes the chosen duration infeasible. A fixed-phase ideal interior satisfies $2q'' = 3\Lambda q^2$, rather than generally $q'' = 0$.

Neither reduced infeasibility nor its optimum is a full Kepler certificate. Constant nonzero acceleration eventually makes $|a_t|r^2$ large. For example, the earlier Earth illustration with 7000 km parking radius, chosen 900000 km SOI and useful $P/m_0 = 540\,\mathrm{W/kg}$ gives an initial minimum powered acceleration $0.00432\,\mathrm{m/s^2}$. Its slow-spiral diagnostic already reaches 0.05 near 67922 km from Earth's center. The calculation is a validity warning for continuous slow spirals, not a prediction of optimized Earth escape [9].

5.2 An exact feasible pair

Exact analytic curves are useful independent tests of a controlled Kepler solver [14]. In local units $\mu = 1$, set $0 \le t \le 10$ and

$$r = 1 + 0.03t^2, \qquad \theta = t/\sqrt{r}, \qquad \theta' = r^{-3/2}, \\ x = re_r, \quad v = 0.06t\,e_r + r^{-1/2}e_\theta, \\ u = 0.06e_r + 0.03t\,r^{-3/2}e_\theta. \tag{68}$$

Direct differentiation gives $x'' = -x/r^3 + u$. It starts on the unit circle, first reaches $r = 4$ at $t = 10$, and ends with $\theta = 5$, $v_r = 0.6$, $v_t = 0.5$, $E = 0.055$. The effort and a full-time acceleration enclosure are

$$U = \frac{651}{16000} + \frac{\pi\sqrt{3}}{240} = 0.0633599920529\ldots, \\ 0.06 \le |u| \le \frac{\sqrt{181}}{150} < 0.09. \tag{69}$$

With $\chi = 0.9$, budget 0.08, floor 0.8 and band $[10, 50]$, $(\chi + U)/50 < 0.06$ and $\chi/10 = 0.09$. Thus the curve is safe and band-feasible throughout, without relying on sampling. It retains mass fraction $0.9342302020\ldots$.

Let $Q = \mathrm{diag}(1, -1)$. The reflected reversal $x_c(t) = Qx(10-t)$, $v_c(t) = -Qv(10-t)$ and $u_c(t) = Qu(10-t)$ preserves prograde angular momentum and arrives at

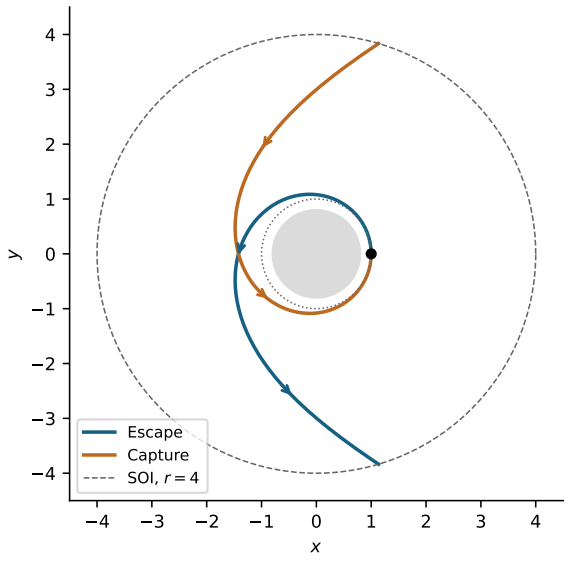


Figure 2: Exact escape and reflected capture examples. The station orbit is dotted, the SOI is dashed, and $r < 0.8$ is excluded. These are separate feasible local trajectories.

$(1, 0, 0, 1)$. A unit station with initial phase -10 rendezvous at this endpoint. The same magnitude enclosure proves band feasibility for the same initial χ, but the mass history is recomputed forward: reversing position and velocity does not reverse fuel use.

For arbitrary endpoint energies in the annulus, define $W_E = \sqrt{2E + 2/\varepsilon}$. Since $|v| \le W_E$, equation (50) gives $|W_E'| \le |u|$; also $|j'| \le \sigma_S |u|$. With $d_* = \max(|\Delta W_E|, |\Delta j|/\sigma_S)$, every full-power/coast transfer satisfies

$$J \ge \max\left\{ \frac{d_*^2}{h},\ \chi\left(e^{d_*/\nu_+} - 1\right) \right\}. \tag{70}$$

The second term follows from $\int |u| dt \le \nu_+ \log(1 + J/\chi)$. It remains valid for nonpositive E whenever the endpoint states are admissible. For (68), the energy part and feasible construction give the symbolic same-endpoint bracket

$$\frac{411 - 30\sqrt{174}}{1000} \le V_{\text{exact}} \le \frac{651}{16000} + \frac{\pi\sqrt{3}}{240}. \tag{71}$$

This establishes a feasible upper bound and a global lower bound in the stated patch model; the constructed curve is not asserted stationary.

6 What establishes optimality

A useful planner needs a feasible trajectory and a reason to trust its reported performance. These are separate outputs. A small shooting residual indicates a stationary candidate; it does not establish a minimum. Comparing feasible branches improves an upper bound, while a lower bound must cover every admissible competitor.

6.1 The smooth second variation

For an exact ideal extremal strictly outside the floor, let $z = (\delta r, \delta v)$ and $\xi = \delta u$ satisfy $z' = Az + B\xi$, $z(0) = z(h) = 0$. The fixed-endpoint second variation is

$$Q[\xi] = \int_0^h \left(2|\xi|^2 - z^T C_s z\right) dt, \tag{72}$$

with $C_s = \operatorname{diag}(K, 0)$ and

$$\begin{aligned} K = \frac{3}{R^5}[p_v r^T + r p_v^T + (p_v \cdot r) I] \\ - \frac{15(p_v \cdot r)}{R^7} r r^T, \qquad R = |r|. \end{aligned} \tag{73}$$

Differentiating $Dg = -I/R^3 + 3rr^T/R^5$ and contracting its derivative with p_v gives the displayed K. The sign and factor two follow by augmenting the minimized cost with $\int p \cdot (s' - f) dt$. Coercivity $Q \ge \gamma_Q \|\xi\|_2^2$ with $\gamma_Q > 0$ proves a strict local minimum in the L^2 control topology; a negative direction on the endpoint kernel rules it out.

Initialize the vertical Jacobi frame $X(0) = 0$, $Y(0) = I$ and integrate

$$\begin{pmatrix} X \\ Y \end{pmatrix}' = \begin{pmatrix} A & D \\ -C_s & -A^T \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}. \tag{74}$$

Conjugate times are rank losses of $X(t)$ at $t > 0$. For this regular fully actuated acceleration problem, absence of conjugate points on $(0, h]$ gives strict local minimality; a first conjugate point strictly before h produces a negative direction on the longer interval [2]. Where X is invertible, $S = YX^{-1}$ satisfies

$$S' + A^T S + SA + SDS + C_s = 0, \tag{75}$$

and completing the square yields $Q = [z^T S z]_0^h + 2\int |\xi - B^T S z/2|^2 dt$. Initialize the frame, not an arbitrary $S(0) = 0$.

Sampling $\det X$ cannot exclude conjugate points: even multiplicity can leave its sign unchanged. The synthetic accessory problem $Q/2 = \int_0^1 (z''^2 - 600 z^2) dt$ in two identical coordinates has such a crossing at $t_c = 0.955712066275\ldots$. It also has the explicit negative direction $z = 1 - \cos(2\pi t)$. The test verifies the diagnostic logic, not a particular orbital branch. Engine switches, radial contacts and flybys need their own junction terms in a second variation.

6.2 Global effort lower bounds

With floor $a > 0$, gravity obeys $|g| \le g_m = a^{-2}$. For the endpoint moments d_r, d_v of (9),

$$L_v = \frac{[|d_v| - g_m h]_+^2}{h}, \qquad L_r = \frac{3[|d_r| - g_m h^2/2]_+^2}{h^3} \tag{76}$$

are lower bounds. They follow by integrating (4) and applying Cauchy–Schwarz. A joint directional bound, for $\lambda = (\alpha, \beta) \ne 0$, is

$$\begin{aligned} L_\lambda = \frac{[\lambda \cdot (d_r, d_v) - g_m \int_0^h |q(t)| dt]_+^2}{\lambda^T W_0 \lambda}, \\ q(t) = (h - t)\alpha + \beta. \end{aligned} \tag{77}$$

An upper enclosure of the convex norm integral preserves this bound. Several directions may be evaluated cheaply at the actual query.

Energy and angular momentum supply additional bounds for competitors with $J \le U$, where $U \ge 0$ is any cutoff. Put

$$\begin{aligned} c_0 &= |v_0|^2 + 2/a - 2/|r_0|, \\ I_v(U) &= [h\sqrt{U} + \sqrt{h^2U + hc_0}]^2, \\ R_U &= |r_0| + h|v_0| + g_m h^2/2 + \sqrt{Uh^3/3}. \end{aligned} \tag{78}$$

To see the first estimate, write $I = \int_0^h |v|^2 dt$. The work identity and Cauchy–Schwarz imply $|v(t)|^2 \le c_0 + 2\sqrt{UI}$, whence $I \le hc_0 + 2h\sqrt{UI}$. Solving this quadratic inequality in $\sqrt{I}$ gives $I_v(U)$. Double integration of acceleration and Cauchy–Schwarz give R_U. Equation (50) gives $\int |v|^2 dt \le I_v(U)$ and $|r| \le R_U$, hence

$$L_E = \frac{(\Delta E)^2}{I_v(U)}, \qquad L_M = \frac{(\Delta j)^2}{hR_U^2}. \tag{79}$$

Degenerate zero numerators and denominators are handled by continuity. A globally valid combination is

$$L = \max\{0, L_v, L_r, L_\lambda, \min[U, \max(L_E, L_M)]\} \le V. \tag{80}$$

The cutoff clipping covers trajectories with effort greater than U. An actually feasible trajectory of effort at most U additionally establishes $V \le U$ and suboptimality at most $U - L$. Ideal lower bounds also apply to the bounded engine at the same endpoints; its upper bound needs a band-feasible witness.

6.3 A global certificate near a low-effort path

Let $(\bar r, \bar v, \bar u)$ be an endpoint-exact feasible ideal reference of effort U and p any continuous piecewise differentiable multiplier. Define residuals

$$e_u = 2\bar u - p_v, \quad e_r = p_r' + Dg(\bar r)^T p_v, \quad e_v = p_v' + p_r. \tag{81}$$

For a competitor, write $\delta u = u - \bar u$, $\delta r = r - \bar r$, $\delta v = v - \bar v$ and $R_g = g(r) - g(\bar r) - Dg(\bar r)\delta r$. Integration by parts gives the exact difference

$$\begin{aligned} J - U = \|\delta u\|_2^2 &+ \int e_u \cdot \delta u \, dt \\ &- \int (e_r \cdot \delta r + e_v \cdot \delta v + p_v \cdot R_g) dt. \end{aligned} \tag{82}$$

Endpoint error would add a boundary term; it cannot be ignored.

Suppose $|\bar r| \ge b > a$, choose $0 < d < b$ and put $\ell = b - d$, $\omega = \sqrt{2/\ell^3}$. Before a first exit from $|\delta r| < d$,

$$|\delta r(t)| \le k_r(t)\|\delta u\|_{2,[0,t]}, \quad |\delta v(t)| \le k_v(t)\|\delta u\|_{2,[0,t]}, \tag{83}$$

where

$$k_r(t)^2 = \int_0^t \frac{\sinh^2(\omega s)}{\omega^2} ds, \quad k_v(t)^2 = \int_0^t \cosh^2(\omega s) ds. \tag{84}$$

Every competitor with $J \le U$ has $\|\delta u\|_2 \le 2\sqrt{U}$. Thus

$$2\sqrt{U}\, k_r(h) < d \tag{85}$$

prevents a first exit and covers all cheaper competitors.

Let $A_r = \int_0^h k_r^2 dt$, $A_v = \int_0^h k_v^2 dt$, $P_v \ge \sup |p_v|$ and

$$\begin{aligned} \gamma &= 1 - \frac{3P_v}{\ell^4} A_r, \\ \eta &= \|e_u\|_2 + \sqrt{A_r}\|e_r\|_2 + \sqrt{A_v}\|e_v\|_2. \end{aligned} \tag{86}$$

The sharp bound is $\|D^2 g(x)[z,z]\| \le 6|z|^2/|x|^4$. Indeed

$$D^2 g(x)[z,z] = \frac{6(x\cdot z)z + 3|z|^2 x}{|x|^5} - \frac{15(x\cdot z)^2 x}{|x|^7}.$$

For $|z| = 1$ and $c = \hat x \cdot z$, its squared norm is $9(1 - 2c^2 + 5c^4)/|x|^8 \le 36/|x|^8$, with equality radially. Taylor's factor $1/2$ therefore gives $|R_g| \le 3|\delta r|^2/\ell^4$. This tightens the earlier factor 12 in γ to 3 and implies $J - U \ge \gamma s^2 - \eta s$ with $s = \|\delta u\|_2$. If (85) holds and $\gamma > 0$,

$$\boxed{\max(0, U - \eta^2/(4\gamma)) \le V \le U.} \tag{87}$$

For an exact extremal $\eta = 0$ proves uniqueness of the global minimum, up to equality of controls almost everywhere. Numerical use requires validated enclosures of the reference, residual norms and constants. Sampled minima with padding do not supply that proof.

For difficult cells a verification function satisfying $W_t + W_r \cdot v + W_v \cdot g - |W_v|^2/4 \ge 0$ throughout every possible cheaper trajectory yields another lower bound by integration. A sampled residual mesh is insufficient.

6.4 The atlas decision

Store model, winding and active-set labels with every branch, together with endpoint residuals, feasible reconstruction, effort enclosure, and the domain and provenance of each optimality claim. A point proof does not certify the surrounding interpolation cell. Uniform continuation, constraint bounds and second-variation or global-gap bounds must cover that cell.

At a query, combine valid lower bounds by their maximum and feasible upper bounds by their minimum. Return infeasible if $L > \kappa$, feasible if $U \le \kappa$, and unresolved otherwise. A failed sufficient test is unresolved. An extra revolution alone is not grounds for rejection: a circular coast over 4π is a two-turn global zero-effort minimum for its own endpoints.

7 Local rendezvous and pursuit

The same effort resource gives an exact local linear problem. In a frame rotating with a reference circle, use $s = (x, y, \dot x, \dot y)$ with unit mean motion. The planar Clohessy–Wiltshire system [4] is

$$\begin{aligned} s' &= As + Bu, \\ A &= \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 3 & 0 & 0 & 2 \\ 0 & 0 & -2 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}. \end{aligned} \tag{88}$$

Let $\Phi(t) = e^{At}$ and $W(h) = \int_0^h \Phi(t)BB^T\Phi(t)^T dt$. Controllability makes W positive definite for every $h > 0$. For $y_f = s_f - \Phi(h)s_0$,

$$J^* = y_f^T W^{-1} y_f, \qquad u^*(t) = B^T\Phi(h-t)^T W^{-1} y_f. \tag{89}$$

The reachable set at budget b is the ellipsoid $J^* \le b$. Projecting to position uses the upper-left 2×2 block of W. The center is the unforced CW trajectory. Appendix E gives the trigonometric transition and Gramian entries. Their Taylor series supply the small-time limit; a scaled Cholesky solve avoids explicit inversion.

7.1 The ideal rendezvous game

Let $z = s_E - s_P$, with reserves b_E, b_P. Set

$$K(t) = \Phi(-t)B, \qquad M(h) = \int_0^h K(t)K(t)^T dt, \qquad C(h) = z_0^T M(h)^{-1} z_0. \tag{90}$$

For $z_0 \neq 0$, C decreases continuously and strictly from infinity to zero. Its derivative is $C'(h) = -|K(h)^T M(h)^{-1} z_0|^2$. The strict monotonicity follows from controllability on every interval; the large-time zero limit follows from the explicit CW Gramian.

Theorem 1 (Ideal CW rendezvous)**.** *With unrestricted L^2 controls, no further state constraints and instantaneous observation of the evader's control by the pursuer, the minimum guaranteed time to full-state rendezvous is the unique $H > 0$ solving*

$$C(H) = (\sqrt{b_P} - \sqrt{b_E})^2 \tag{91}$$

when $b_P > b_E$. If $b_P \le b_E$ and $z_0 \neq 0$, the value is infinite. At $z_0 = 0$ it is zero.

For the upper bound use $a_H(t) = K(t)^T M(H)^{-1} z_0$ and $u_P = u_E + a_H$. The relative endpoint is exactly zero, and Minkowski's inequality gives $\|u_P\|_2 \le \sqrt{b_E} + \sqrt{C(H)} = \sqrt{b_P}$. The opposing strategy and parity argument appear in Appendix B; the result uses the integral-constraint game framework [7]. Full-state rendezvous is not position-only interception or capture within a finite radius.

7.2 A sufficient nonlinear extension

For rotating position q, the exact Kepler equation is

$$q'' + 2\mathsf{J}q' = \mathrm{diag}(3,0)q + u + N(q), \tag{92}$$

where $N(q) = -(e_1 + q)/|e_1 + q|^3 + e_1 - \mathrm{diag}(2,-1)q$. On $|q| \le \epsilon < 1$,

$$|N(q)| \le N_\epsilon = \frac{3\epsilon^2}{(1-\epsilon)^4}, \quad \mathrm{Lip}(N) \le L_\epsilon = \frac{6\epsilon}{(1-\epsilon)^4}. \tag{93}$$

These finite-domain inequalities follow from $\|D^2 g(x)[u,v]\| \le 6|u||v|/|x|^4$ and Taylor's formula.

Use $u_P = u_E + a_h + N(q_E) - N(q_P)$. It cancels the nonlinear relative remainder, giving exactly the CW correction trajectory z_h. If both paths remain in the ball, then

$$\sqrt{b_P} \ge \sqrt{b_E} + \sqrt{C(h)} + L_\epsilon \left(\int_0^h |\Pi z_h(t)|^2 dt \right)^{1/2} \tag{94}$$

guarantees rendezvous by h, where $\Pi = (I, 0)$. Appendix C gives an explicit first-exit tube bound, and Appendix D gives an inertial half-space alternative for wider separations. The same tube inflates a single-ship reachable ellipsoid to budget $(\sqrt{b} + N_\epsilon\sqrt{h})^2$, giving an outer reachability bound.

These capture results concern the ideal engine. A restricted ship's reachable set is contained in the ideal set, but restricting both players need not preserve the ideal winner. A bounded-engine capture guarantee requires admissibility against every permitted opponent. Checking one saddle trajectory or clipping the compensating control does not establish that guarantee.

8 Interception and deadline-fuel games

This section restores position-only interception, needed for contact or salvage races. Boarding instead requires velocity matching. All closed forms here use the ideal engine and instantaneous nonanticipative response; they are not finite-band game values. Write $r = x_E - x_P$, $w = v_E - v_P$, and take capture to mean $|r| \le R$. Resources in this section may be physical, with $g = -\mu x/|x|^3$, or consistently normalized.

8.1 The four problems and verification

Let $z = (x_P, v_P, x_E, v_E, b_P, b_E)$ and

$$\mathcal{D}U = U_{x_P} \cdot v_P + U_{v_P} \cdot g(x_P) + U_{x_E} \cdot v_E + U_{v_E} \cdot g(x_E).$$

The Isaacs operator for pursuer running cost $\ell|a_P|^2$ is

$$\mathcal{H}_\ell U = \mathcal{D}U + \inf_{a_P}\{U_{v_P} \cdot a_P + (\ell - U_{b_P})|a_P|^2\} + \sup_{a_E}\{U_{v_E} \cdot a_E - U_{b_E}|a_E|^2\}. \tag{95}$$

At a zero resource face the corresponding control is zero. Where $c_P = \ell - U_{b_P} > 0$ and $c_E = U_{b_E} > 0$, this becomes

$$\mathcal{H}_\ell U = \mathcal{D}U - \frac{|U_{v_P}|^2}{4c_P} + \frac{|U_{v_E}|^2}{4c_E}, \qquad a_P^* = -\frac{U_{v_P}}{2c_P}, \qquad a_E^* = \frac{U_{v_E}}{2c_E}. \tag{96}$$

Zero or wrong-sign coefficients require evaluating the original extrema, including infinite values, rather than dividing by zero. Problems A1 and B1 minimize guaranteed interception time in free space and Kepler gravity, respectively: $1 + \mathcal{H}_0 V = 0$, $V = 0$ on capture. Problems A2 and B2 minimize worst-case pursuer effort to capture within remaining time s: $W_s = \mathcal{H}_1 W$, with $W = 0$ on capture and $W(0, z) = +\infty$ outside capture. Failure to capture by the deadline costs $+\infty$; the budget state remains a hard constraint.

Proposition 2 (Classical verification)**.** *Suppose a nonnegative differentiable solution satisfies the stated PDE, target and resource boundary conditions. Suppose its selectors generate well-posed admissible responses throughout the relevant domain. For time, the pursuer selector guarantees capture and the evader selector gives the opposing time bound. For deadline effort, assume additionally that the pursuer selector guarantees capture before the deadline. Then the solution is the respective value, provided the boundary limits used at capture and exhaustion exist.*

Indeed, along the time pursuer selector, $dV/dt \le -1$ against every evader, whereas along the evader selector $dV/dt \ge -1$ against every pursuer. Integrate to capture; nonnegativity supplies the time upper bound. For remaining time $s' = -1$, along the deadline pursuer selector, $dW/dt + |a_P|^2 \le 0$, and along the evader selector the inequality reverses. Integration gives the opposing effort bounds; noncapture has infinite cost. Localization and boundary limits are essential. Nonsmooth values require viscosity and comparison hypotheses [5]; this verification statement does not assert existence for an unrestricted singular orbital domain.

8.2 Closed forms and distinctions

Against a ballistic evader in free space, set $d(h) = (1 - R/|r + hw|)_+(r + hw)$, with $d = 0$ at zero separation. The exact prescribed-time interception effort and control are

$$c(h) = \frac{3|d(h)|^2}{h^3}, \qquad a_P(t) = \frac{3(h-t)}{h^3} d(h). \tag{97}$$

These follow by projecting the required position onto the capture ball and using the single moment $\int_0^h (h - t) a_P dt = d$. Deadline fuel is $\inf_{0<h\le T} c(h)$ and earliest interception is $\inf\{h : c(h) \le b_P\}$. In particular, if $c(h) > b_P$ for every $0 < h \le T$, coasting is a zero-fuel evader win, even when fuel is available.

For active interception with $w = 0$, $D = |r| - R > 0$, $b_P > b_E$ and $b_P \ge 4b_E$, the exact guaranteed time is

$$H_I = \left[\frac{3D^2}{(\sqrt{b_P} - \sqrt{b_E})^2} \right]^{1/3}. \tag{98}$$

Put $k = \sqrt{b_E}/(\sqrt{b_P} - \sqrt{b_E}) \le 1$ and $a_0(t) = 3D(1 - t/H_I) r/(H_I^2 |r|)$. The response $a_P = a_E + a_0$ gives the upper bound by Minkowski. The opponent $a_E = k a_0$ prevents an earlier hit: at $t = sH_I < H_I$ any pursuer needs at least $Cs^{-3}[1 + kf(s)]^2$, where $C = 3D^2/H_I^3$ and $f(s) = (3s^2 - s^3)/2$. This exceeds $(1 + k)^2 C = b_P$ because

$$1 + kf(s) - (1 + k)s^{3/2} = (1 - k)(1 - s^{3/2}) + k[1 + f(s) - 2s^{3/2}] > 0;$$

use $f(s) \ge s^3$. For point capture ($R = 0$), full-state boarding takes $4^{1/3} H_I$ under these same hypotheses.

For a prescribed exact endpoint at time h, let

$$\begin{aligned} C_I &= 3|r + hw|^2/h^3, \\ C_J &= 12|r|^2/h^3 + 12 r \cdot w/h^2 + 4|w|^2/h. \end{aligned} \tag{99}$$

The endpoint minimax pursuer effort is

$$M_{I,J}(h) = (\sqrt{b_E} + \sqrt{C_{I,J}(h)})^2. \tag{100}$$

The copying response gives the upper bound. Choosing the evader control proportional to the minimum-norm endpoint correction gives the lower bound by orthogonal projection onto the endpoint moment constraints. This is an exact-time endpoint result. Minimizing it over deadlines gives a sufficient first-hit bound, not automatically the first-hit value. For perpetual avoidance of exact rendezvous at a nonzero phase state, $b_E \ge b_P$ gives least worst-case evader expenditure $F_\infty^J = b_P$. The parity feedback in Appendix B attains the upper bound; a smaller uniform expenditure cap would give the pursuer a strictly positive resource gap and finite-time rendezvous.

Nonattainment matters. Strict position-only escape can have an infimum without a minimizing control. An exhausted stationary pursuer at the origin and an evader coasting from $(-L, R)$ with velocity $(u, 0)$ meet the capture boundary at L/u. Any arbitrarily small positive transverse thrust before that time avoids it forever, at cost tending to zero; zero thrust loses. There are also positive unattained infima. In one dimension take point capture, separation $L > 0$, closing velocity $-u < 0$, and an exhausted pursuer. The critical effort is $4u^3/(9L)$: the endpoint lower bound $3(ut - L)^2/t^3$ is largest at $t = 3L/u$. Equality forces contact there. The safe profiles $r(t) = \delta + (L - \delta)(1 - t/t_\delta)^3$ for $0 \le t \le t_\delta = 3(L - \delta)/u$, followed by rest at $\delta > 0$, approach that cost as $\delta \downarrow 0$ without attaining it.

8.3 Five-state reduction and scaling

In isotropic free space a value depends only on $(\rho_G, \xi_G, \omega_G, b_P, b_E) = (|r|^2, r \cdot w, |w|^2, b_P, b_E)$, where $\xi_G^2 \le \rho_G \omega_G$. The drift is $2\xi_G U_{\rho_G} + \omega_G U_{\xi_G}$ and

$$|U_w|^2 = \rho_G U_{\xi_G}^2 + 4\xi_G U_{\xi_G} U_{\omega_G} + 4\omega_G U_{\omega_G}^2. \tag{101}$$

Both controlled velocity gradients have this squared norm. These chain rules reduce the free-space Isaacs equation to five states in $d = 2$ or 3. Under $t \mapsto \alpha t$, $r \mapsto \beta r$, $w \mapsto (\beta/\alpha) w$, resources scale as β^2/α^3. Time values scale by α, and deadline-fuel values by β^2/α^3, with radius and deadline scaled as well. Scaling every strategy and opponent proves the rule for values, not only paths.

Finally, restore dimensional mean motion n in the CW matrices. As $n \to 0$, Φ_n and W_n converge to the double-integrator matrices. Theorem 1 therefore reduces to the free-space cubic

$$(\sqrt{b_P} - \sqrt{b_E})^2 H^3 - 4|w|^2 H^2 - 12 r \cdot w H - 12|r|^2 = 0. \tag{102}$$

One takes this physical limit before setting the normalized mean motion to one.

9 Heliocentric checks and worked bounded cases

Use $R_0 = 149597870700$ m and $\mu_\odot = 1.32712440018\,10^{20}$ m^3/s^2. Then $T_0 = 58.13244087$ days, $V_0 = 29.78469183$ km/s and $J_0 = 176.6257101$ m^2/s^3.

9.1 Revisiting the Jupiter-class candidates of the transfer article

The ideal engine of [12] has $P = 1$ GW, $m_0 = 3\,10^6$ kg and $m_d = 10^6$ kg: $\chi = 3.774459936$ and $\kappa = 7.548919872$. The public atlas of [12] supplies warm starts, not certified trajectories. We independently integrate the normalized Cartesian state, effort adjoints and effort quadrature, and solve four endpoint equations plus $J = \kappa$. This avoids relying on the repository viewer's radius and angle acceptance tolerances. The geometries below reproduce the short-arc table and the two illustrated solar-pass families of the transfer article, allowing for its rounded figure labels.

ρ, Θ	days	$r_{\min}$ AU	$v_{e,\min}$	$v_{e,\max}$
$5.2, 80°$	170.034	1.0000	40.60	2984.8
$5.0, 273.1°$	206.560	0.16993	42.35	13001.0
$5.2, 385.2°$	214.236	0.000273	4.85	1203.5

Exhaust speeds are in km/s. All three paths fail the band. The first two have numerical radius minima above 0.1 AU; the last also fails the floor. Scaled endpoint residuals are below $4\,10^{-13}$ for the first two and 10^{-8} for the last. The long-arc solve uses polar equations and analytic shooting sensitivities to preserve its winding; a radial-velocity root refines its narrow perihelion, which uniform samples alone miss.

We also reconstruct all 73 rows of that article's 5.2 AU final-polar-angle table, $\Theta = 0°, 5°, \ldots, 360°$, by independent continuation. Every row violates the exhaust band. Their reconstructed times range from 169.7643 to 220.4991 days (the 0° row lies outside the abbreviated 170–215 day range). Scaled endpoint residuals are below $4.8\,10^{-11}$. The radial audit locates turning points rather than only testing a time grid:

Final-angle rows	Band rejected	Floor rejected
0°–300° (61 rows)	61	0
305°–360° (12 rows)	12	12

For example, the numerical minimum radius is 0.1011887 AU at 300° and 0.090577 AU at 305°. Thus none of these 73 candidates, nor either illustrated solar-pass candidate, survives both operational checks.

The audit evaluates $v_e = V_0(\chi + I)/|u|$ and the radius on a dense 40001-point reconstruction, with continuous refinement of the long-arc perihelion and exhaust extrema. Each reported band rejection has a large violation on a continuous powered interval, so it does not depend on interpreting an isolated zero-thrust sample. Clipping a failed trajectory to the band does not repair its endpoint. The input seeds and normalized initial covectors are provided as supplementary material.

9.2 A local certificate demonstration: a 2.9-day solar hop

This example raises the heliocentric radius from 1 to 1.001 AU in 2.90663 days and ends on a slightly eccentric orbit. Its purpose is to demonstrate an explicit feasible path together with global effort and time bounds on a small, fully checkable case. For this auditable local example, keep $P = 1$ GW and $m_0 = 3\,10^6$ kg, but choose the smaller resource $\kappa = 0.0320066666667$. Begin on the unit circular orbit and let $c = 0.4$, $H = 0.05$. Define a physical heliocentric path through its normalized polar coordinates:

$$r(t) = 1 + ct^2, \qquad \theta(t) = \frac{t}{\sqrt{1+ct^2}}, \qquad v_r = 2ct, \qquad v_\theta = r^{-1/2}. \tag{103}$$

Since $\theta' = r^{-3/2}$, exact substitution into the Kepler equations gives $u_r = 2c$ and $u_\theta = ct/r^{3/2}$. Thus this is a continuously powered exact path with $r \geq 1$ AU, not a patched approximation. Its endpoint defines the target: take the unforced Kepler orbit through $(r(H), \theta(H), v_r(H), v_\theta(H))$ at epoch H. The arrival orbit is slightly eccentric. Coasting along that orbit is admissible on the short horizon used below. The same backward comparison used below bounds its separation from the unit circle by 0.0031 on $[0, H]$, so the target orbit also clears the floor.

The effort and a convenient elementary upper bound are

$$\begin{aligned} J &= 4c^2H + c^2\int_0^H \frac{t^2}{(1+ct^2)^3}dt \\ &= 0.0320066546838, \\ U &= 4c^2H + c^2H^3/3 = 0.0320066666667 = \kappa. \end{aligned} \tag{104}$$

Mass accounting gives $m_d = 2974774.5061$ kg from the budget and an actual propellant expenditure 25225.4845 kg for this witness. The burn satisfies $0.8 \leq |u| \leq \sqrt{0.64 + 0.16H^2}$ and $\chi \leq w \leq \chi + U$. Consequently its full-power exhaust remains between 140.4825 and 141.7181 km/s, strictly within the required band, throughout the path.

Here is a global, conservative lower-bound calculation, which also brackets the minimum time instead of assigning optimality to this shaped path. Compare every competitor of effort at most U with the unit circular coast. In a position tube of radius $d = 0.1$, the gravity Lipschitz constant is $K = 2/0.9^3$, $\omega = \sqrt{K}$. Variation of constants and Cauchy–Schwarz give

$$|\delta r(t)| \leq k_r(t)\sqrt{J}, \qquad |\delta v(t)| \leq k_v(t)\sqrt{J}, \tag{105}$$

where k_r, k_v are those in Section 6. Since $\sqrt{U}k_r(H) < 0.0012 < 0.1$, the first-exit argument validates this tube for every cheaper competitor, irrespective of its thrust direction. The target endpoint has $|\delta v(H)| > 0.0399$ and $k_v(H)^2 \leq H\cosh^2(\omega H) < 1.007H$. Therefore

$$0.0316 = L \leq F^*(0, H) \leq U = 0.0320066666667. \tag{106}$$

The clipping argument from (80) covers more expensive competitors as well. These same bounds hold for the

restricted engine, because its witness passes the band and its competitors are a subset of the ideal ones.

For an earlier arrival at $h_- = 0.049$, compare the ballistic target to the unit coast backward from H. At H, $|\delta r| < 0.00101$ and $|\delta v| < 0.041$. Over $H - h_- = 0.001$, the comparison inequalities give $|\delta r| < 0.0011$ and a change in δv of less than $K(0.0011)(0.001) < 0.0000031$. Hence every affordable earlier competitor would need

$$J \geq \frac{(0.0399 - 0.0000031)^2}{1.007(0.049)} > 0.0322591 > \kappa, \quad (107)$$

a contradiction. Coast extension and the monotone value then exclude all still earlier arrivals. Thus the resource-to-time-to-fuel calculation is

$$\boxed{\begin{array}{c} 0.049 < h^*(\kappa) \leq 0.05, \\ 2.84848 < T^* \leq 2.90663 \text{ days.} \end{array}} \quad (108)$$

The stated fuel is that of the witness path; the optimal duration lies somewhere in this interval. The conservative decimal inequalities can be checked by rational Taylor enclosures; a supplementary checker does so for the bounds used in the time bracket.

9.3 A certified 70-day Earth–Mars witness

For an interplanetary performance point, return to $P = 1$ GW, $m_0 = 3\,10^6$ kg and $m_d = 10^6$ kg, hence $\kappa = 7.548919872$. Use coplanar circular planetary orbits at 1 and $R_M = 1.524$ AU in the solar two-body model. The following explicit path gives full position-and-velocity rendezvous in exactly 70 days, with useful jet power fixed at 1 GW throughout. The exhaust schedule is variable and satisfies the two-sided 50–250 km/s band.

Let $H = 70(86400)/T_0$ and $q = 0.51011006$, with the decimal exact. Set $c_q = (1 - q^2)/(1 + q^2)$ and $s_q = 2q/(1 + q^2)$. The endpoint states are

$$\begin{aligned} r_0 &= (1, 0), & v_0 &= (0, 1), \\ r_8 &= R_M(c_q, s_q), & v_8 &= R_M^{-1/2}(-s_q, c_q). \end{aligned} \quad (109)$$

The circular target passes through this endpoint at H. Thus its initial phase is $2\arctan q - H/R_M^{3/2}$, approximately 17.3820101° ahead of Earth; arrival longitude is approximately 54.0531714°. Planetary escape, capture, parking-orbit costs and departure waiting time are outside this heliocentric model.

On interval $j = 0, \ldots, 7$, put $\Delta_j = t_{j+1} - t_j$, $s = (t - t_j)/\Delta_j$ and use cubic Hermite interpolation:

$$\begin{aligned} r(t) = (2s^3 - 3s^2 + 1)r_j + (s^3 - 2s^2 + s)\Delta_j v_j \\ + (-2s^3 + 3s^2)r_{j+1} + (s^3 - s^2)\Delta_j v_{j+1}. \end{aligned} \quad (110)$$

Table 1 and the analytic endpoints specify the whole path. Define its control and mass by

$$\begin{aligned} u &= r'' + r/|r|^3, & I(t) &= \int_0^t |u|^2\, ds, \\ m(t) &= \frac{m_0\chi}{\chi + I(t)}, & v_e(t) &= V_0\frac{\chi + I(t)}{|u(t)|}. \end{aligned} \quad (111)$$

The position and velocity are continuous and the endpoint conditions hold exactly by construction. Acceleration may jump at knots; the model imposes no thrust-slew constraint. Each one-sided control value passes the band. The resulting mass, thrust and exhaust obey the constant-power equations exactly almost everywhere.

A whole-interval enclosure gives

$$\begin{array}{c} 2.5212 < J < 2.5307 < U_M = 2.6 < \kappa, \\ 99 < v_e(t) < 102 \text{ km/s}, \qquad |r(t)| > 0.999 \text{ AU}, \\ 1.2014\,10^6 < m_0 - m(H) < 1.2041\,10^6 \text{ kg}. \end{array} \quad (112)$$

In particular, the spacecraft retains over $1.7959\,10^6$ kg, above the dry-mass floor. Consequently, for this target phase,

$$\boxed{F^*_{\text{band}}(0, H) \leq U_M = 2.6, \qquad T^*_{\text{band}}(\kappa) \leq 70 \text{ days.}} \quad (113)$$

This is an unoptimized upper bound on the minimum time. Comparing it with an ideal-engine optimum for the same endpoints, to measure the time cost of the band, is left for future work.

Here is a reproducible enclosure prescription. Form the Hermite cubic coefficients from the defining decimals, using outward interval arithmetic at 60 significant decimal digits, including the square roots in T_0, V_0 and the terminal velocity. Subdivide each of the eight local s intervals into 128 equal parts. Horner evaluation encloses r and r''; square, sum and take an outward square root to bound $|r|$, then enclose u and $|u|^2$. Multiplying each latter enclosure by its time width gives lower and upper Riemann bounds for I. For the exhaust check on a subinterval, use the cumulative lower bound at its start and cumulative upper bound at its end. This monotonic enclosure covers every time within it. The resulting bounds imply the displayed coarse inequalities. A standalone supplementary checker implements this recipe; a separate floating-point quadrature and dynamics integration provide independent numerical cross-checks.

10 Computation and reproducible evidence

10.1 Offline branches, online decisions

The intended runtime is a few milliseconds, so full boundary-value solves belong offline. Continue endpoint-exact branches over geometry, time, vehicle and active-set coordinates; retain competing windings and encounter sequences. Audit engine and radial feasibility, compute variational data, and attach the strongest verified performance bounds available. A surrogate must preserve its branch domain and an error bound; interpolated controls need endpoint correction.

At runtime, normalize the actual vehicle and endpoints, retrieve candidate branches, and propagate actual mass through the route. Use (44) or (54) to optimize stored phase couplings, evaluate lower and feasible upper bounds, and search duration with a maintained bracket. Interior stationary events disappear from the

Table 1: Exact rational interior knots of the 70-day witness. Entries are exact defining decimals. Positions and velocities use the solar normalized units. Endpoint knots are $t_0 = 0$, $t_8 = H$ and the analytic states in the text.

j	t_j/H	r_{jx}	r_{jy}	v_{jx}	v_{jy}
1	0.137169299389	1.001350238628	0.168727459300	0.017525621592	1.046344765303
2	0.274338598779	1.006411818893	0.346210491047	0.046245191438	1.105077859977
3	0.411507898168	1.017566072445	0.534820743099	0.091838207401	1.182668462417
4	0.548677197557	1.037723200329	0.738442255030	0.154970973170	1.288219089473
5	0.661507898168	1.043707872696	0.902107682569	-0.064178195495	1.117609239693
6	0.774338598779	1.020891188142	1.041194789559	-0.269833623616	0.926300052106
7	0.887169299389	0.970819319098	1.152810168550	-0.465866897104	0.712855383845

outer first variation, but their feasibility and optimality status remain part of the route status. An unresolved budget test calls for refinement, not an infeasibility label.

The following distinctions are useful output fields:

Status	Evidence required
Feasible	Endpoint, dynamics, resource and path checks
Stationary	Correct PMP and all junction residuals
Local minimum	Valid second-order sufficiency
Bounded gap	Valid global L and feasible U
Global minimum	Matching bounds or a global proof

These fields are not inferred from a solver's successful return code. Normal PMP conditions alone also do not enumerate possible abnormal extremals or prove ordinary-control attainment in a nonconvex on/off problem.

10.2 Three consistency checks

The following reproducible identities check the interfaces between formulas.

Timing and control gain. Circular-endpoint shooting checks both $F_h = -|u_f|^2$ and $F_d = (\omega - 1)\mathcal{L}$ against its endpoint data. Substituting $p_v = 2u$, $C = 1$ in the bounded selector recovers $\mathcal{G} = |u|^2$ whenever the band is inactive. The exact-rational nonconvexity certificate is rerun separately; it proves the leading Jensen-gap sign, not global optimality of a selected finite perturbation.

One physical mass, several local units. Take an artificial accounting example with $m_0 = 1000$ kg, $P = 10^6$ W, and two physical efforts $j_1 = 20$, $j_2 = 40$ in $\mathrm{m}^2/\mathrm{s}^3$. Equation (59) gives $m_f = 970.8737864078$ kg. Assigning effort scales $J_{0,1} = 2$ and $J_{0,2} = 5$ changes the normalized efforts to 10 and 8, but gives the same mass when each entry χ is computed from the actual entry mass. Incorrectly resetting the second leg to m_0 gives 980.3921568627 kg. With the second power changed to $2\,10^6$ W, the correct final mass is also 980.3921568627 kg: the equality is coincidental and illustrates why power belongs in the accounting. These are algebraic examples, not trajectories.

SOI action and covectors. Deterministic randomized transverse crossings verify that (53) makes all four chart derivatives in (54) vanish, that the relative Hamiltonian matches, and that the state saltation matrix produces the same covector jump. Independent scale-pairing checks verify (58). Both entry and exit signs are exercised. These are interface identities; they do not demonstrate an optimized complete route.

10.3 Trajectory test cases

The reduced spiral test cases include infeasibility at $h = 480$ and efforts 0.00025 at $h = 1000$ and 0.000100501670842 at $h = 3000$, for the parameters of Figure 1. Independent quadrature and time-domain integration check the scalar formulas. The exact escape/capture pair in Figure 2 has an analytic dynamics and band proof; vector ODE integration independently checks its endpoints. A powered-flyby test case verifies work, mass and asymptotic labels on a feasible finite-SOI pass.

The exact-radius and Jensen calculations use rational arithmetic with explicit enclosures. Shooting, transcription and trajectory audits use double precision. The input data, numerical tolerances and scripts are provided as supplementary material.

A benchmark of the scalar circular-profile kernel gave a median of about 0.071 ms over 300 C++ queries. It measures only that reduced scalar kernel. It includes neither full orbit integration nor constrained boundary-value optimization and gives no complete-route latency guarantee.

11 The remaining computational problem

The derivations provide a consistent chain from physical power and mass to fixed-time actions, timing gradients, bounded controls, safety contacts and planetary interfaces. Exact local models and global inequalities can reject some queries or bound their cost without solving a new transfer. This article assembles those results in one convention and includes its analytical certificates as appendices.

The next substantial computational advance is an endpoint-exact, finite-band, safety-constrained trajectory atlas with validated cells and general encounter velocities. A complete station-to-station demonstration should then join escape, heliocentric flight and capture using the actual mass and station phases, compare com-

peting branches, and report a feasible cost together with its lower bound. Constrained second variations, finite-pulse realization, and model error relative to three-body dynamics remain distinct extensions. The framework specifies these tasks without treating them as already solved.

A Capped free-space propagation

For the dimensional time problem in one dimension, let $l = \lambda_v$ with $l' = -\lambda_x$ and λ_x constant. On a powered fixed-sign interval, write $|l| = L_a + nt$. For fixed exhaust V set $F = 2P/V$, $\sigma = 2P/V^2$ and $C_m = \lambda_m m^2$. Differentiating the optimized Hamiltonian gives

$$m' = -\sigma, \quad \lambda_m' = \frac{F|l|}{m^2}, \quad C_m' = \frac{2P|l|}{V} - \frac{4PC_m}{mV^2}. \tag{114}$$

Thus C_m is not constant on this boundary. With $B = n/\sigma$ and $A = L_a + Bm_a$,

$$\begin{aligned} m &= m_a - \sigma t, \\ \lambda_m &= \lambda_{m,a} + V\left[A\left(\frac{1}{m} - \frac{1}{m_a}\right) + B\log\frac{m}{m_a}\right]. \end{aligned} \tag{115}$$

For thrust sign $\epsilon = \pm 1$,

$$\begin{aligned} v &= v_a + \epsilon V \log(m_a/m), \\ x &= x_a + v_a t + \epsilon V\left[t - \frac{m}{\sigma}\log(m_a/m)\right]. \end{aligned} \tag{116}$$

On an ideal interior arc C_m is constant and $a = Pl/C_m$; on a coast both m and λ_m are constant.

A.1 Relations carried over from the uncapped analysis

The pointwise selector holds when evaluated with the current C_m. The fixed-exhaust state maps above, the interior-arc mass formula, and the exchange argument excluding an avoidable powered–coast rearrangement under its monotone braking hypotheses also hold. On an interior arc $l = l_a - nt$,

$$\frac{1}{m(t)} = \frac{1}{m_a} + \frac{P}{2C_m^2}(l_a^2 t - l_a n t^2 + n^2 t^3/3). \tag{117}$$

Integrating $a = Pl/C_m$ gives quadratic velocity and cubic position. The raw primer remains affine globally; its scaled acceleration does not. Because C_m varies on boundary arcs, switching times cannot be obtained from a constant-C_m primer, not even with its terminal value.

The exchange argument uses states only. In a coast/ceiling-brake mixture with duty fraction $\delta \in [0,1]$, write $s(t) = \int \delta\, dt$. Then $m = m_a - \sigma s$ and $v = v_a - V\log(m_a/m)$. Final mass and velocity depend only on total powered time. Exchanging a burn followed by a coast of length c for coast followed by burn increases distance by $c\,\Delta v > 0$, where Δv is the burn's velocity decrement. Repetition puts all coasting first. If the braking interval has $0 \le v \le v_a$ and $v_a > 0$, its original coast distance is reproduced by a shorter leading coast of duration $\int_{\text{coast}} v\, dt/v_a$. Mass, terminal velocity and distance are unchanged, while elapsed time strictly decreases whenever a positive coast follows a positive brake. Approximation extends this state argument to fractional mixtures.

A.2 Ceiling-only topology and the unresolved two-sided case

Consider the ceiling-only, positive-distance, rest-to-rest problem with $C_m > 0$ and positive dry mass. Normalize the constant $\lambda_x = n > 0$; the primer is $l = n(t_z - t)$. Define $z = Vm|l|/(2C_m)$. The selector gives interior U for $z > 1$, ceiling burn B for $1/2 < z < 1$, and coast C for $z < 1/2$. On a ceiling arc,

$$\frac{z'}{z} = \frac{|l|'}{|l|} + \frac{\sigma}{m}(1 - 2z), \qquad \sigma = 2P/V^2. \tag{118}$$

On an interior arc $z'/z = |l|'/|l| - \sigma z^2/m$; on a coast it is $|l|'/|l|$. During boost $|l|' < 0$, so z decreases strictly through $\mathsf{U}_+, \mathsf{B}_+, \mathsf{C}$, with initial truncations allowed. On braking $|l|' = n > 0$. At $z = 1/2$ both the coast and boundary vector fields have $z' > 0$. Thus $\mathsf{C} \to \mathsf{B}_-$ is regular and cannot return to the tie: every encounter with that level would point upward. No sliding arc or terminal m_6 event is available.

At $z = 1$ both powered fields give $z'/z = n/|l| - \sigma/m$. The quantity $k = mn/(\sigma|l|)$ decreases strictly throughout braking, since m decreases and $|l|$ increases. An upward crossing $\mathsf{B}_- \to \mathsf{U}_-$ can occur only while $k > 1$, and a subsequent downward crossing $\mathsf{U}_- \to \mathsf{B}_-$ only when $k < 1$; no further reversal is possible. Consequently every such finite-switch normal candidate has the ordered pattern

$$\mathsf{U}_+\ \mathsf{B}_+\ \mathsf{C}\ \mathsf{B}_-\ \mathsf{U}_-\ \mathsf{B}_-, \tag{119}$$

with absent intervals and endpoint truncations permitted. In particular the last boundary burn ends at the endpoint, not at a return to coast. This is a ceiling-only one-dimensional result. It does not classify the game's two-sided 50–250 km/s engine. In particular, the 50 km/s lower exhaust limit caps acceleration and introduces additional arcs.

To state precisely what remains unresolved here, write $V_- = 50$ km/s, $V_+ = 250$ km/s, $\rho_V = V_+/V_- = 5$, and define z with $V = V_+$. For $C_m > 0$, the dimensional selector becomes

$$|a| = \begin{cases} 0, & z < 1/2, \\ 2P/(mV_+), & 1/2 < z < 1, \\ P|l|/C_m, & 1 < z < \rho_V, \\ 2P/(mV_-), & z > \rho_V. \end{cases} \tag{120}$$

The last region is a maximum-acceleration (a_+) arc, denoted $\mathsf{A}_\pm$ according to thrust sign. Its state and mass-adjoint maps are the fixed-exhaust maps above with $V = V_-$, and

$$\frac{z'}{z} = \frac{|l|'}{|l|} + \frac{\sigma_-}{m}\left(1 - \frac{2z}{\rho_V}\right), \qquad \sigma_- = 2P/V_-^2. \tag{121}$$

Thus the selector and propagation are available for both boundaries, but a complete two-sided switching-topology theorem, including all $\mathsf{A}_\pm$ visits and admissible endpoint truncations, is left open, as are gravity, abnormal extremals and other endpoint conditions.

A.3 Endpoint reduction

This reduction and its numerical example retain the ceiling-only assumptions of the proved topology. Use homogeneity to fix $n = 1$. The two shooting unknowns are t_z and $\lambda_m(0) > 0$. Propagate the exact interior, boundary and coast maps, switching with the current z, until the first exhaustion $m = m_d$. Call that time $T(t_z, \lambda_m(0))$. Solve

$$x(T; t_z, \lambda_m(0)) = L, \qquad v(T; t_z, \lambda_m(0)) = 0. \quad (122)$$

Reject trajectories without exhaustion or with invalid adjoint sign or switch inequalities. Alternatively use T as a third unknown and add $m(T) = m_d$ as a third residual. Normalize the free-time Hamiltonian afterward if a positive normal scaling exists; and compare competing solutions, since the reduction does not imply uniqueness.

For $P = 1/2$, $V = 2$, $m_0 = 1$, $m_d = 1/2$, $L = 1$, independent numerical integration of the selector gives $t_z = 1.22365494$, $\lambda_m(0) = 0.61524660$, $T = 2.35199587$; the maximum endpoint residual is $9.6\,10^{-8}$. The coast is approximately $[0.80144, 1.64581]$. Direct transcription with piecewise-constant acceleration, exact $(x, v, 1/m)$ propagation and the ceiling imposed at each burn cell's final mass gives:

Burn cells	Time	Endpoint error
12	2.35489579	$2.6\,10^{-12}$
24	2.35338380	$1.4\,10^{-12}$
48	2.35267314	$1.3\,10^{-12}$

The within-cell ceiling holds because inverse mass increases monotonically and the acceleration is constant. Ceiling violations at optimizer tolerance are below $9\,10^{-13}$. The transcription independently checks the trajectory equations and approaches the shooting time from above.

B The opposing CW strategy

For $b_P > b_E$ let H solve the game equation and put $k = \sqrt{b_E}/(\sqrt{b_P} - \sqrt{b_E})$. The evader starts with $u_E = k a_H$. Any rendezvous at $t < H$ needs relative correction a_t. Minimum-norm orthogonality and $\int_0^t a_H \cdot a_t \, ds = C(H)$ imply

$$\int_0^t (|u_P|^2 - |u_E|^2) ds \geq C(t) + 2kC(H) > b_P(0) - b_E(0). \quad (123)$$

Rendezvous cannot occur before, or at, the first resource parity. At parity the evader switches to the following feedback, which also handles an initial $b_E \geq b_P$.

Reset the clock at a nonzero state z_* and let $l = z_*/|z_*|$. Introduce virtual evader reserve $q(0) = b_P(0)$, $q' = -|u_E|^2$ and define

$$Y = l^T \Phi(-t) z, \quad c = B^T \Phi(-t)^T l, \quad \delta = q - b_P, \quad u_E = (\delta/Y) c. \quad (124)$$

Initially $Y > 0$, $\delta = 0$. While $Y > 0$, $\delta' = |u_P|^2 - |u_E|^2$, so $\delta \geq 0$, and

$$(\delta Y^2)' = Y^2 |u_P - u_E|^2 \geq 0. \quad (125)$$

If the product becomes positive, $\delta \leq q(0)$ prevents any finite zero of Y. If it stays zero, both controls vanish almost everywhere and Y stays constant. Thus rendezvous is prevented and the virtual budget bounds actual evader expenditure. The zero-reserve case is coasting. This completes the opposing bound under the theorem's information convention.

C An explicit nonlinear capture tube

To make (94) constructive, define

$$\begin{aligned} g_h &= h + h^2 + \tfrac{2}{3}h^3, \\ G_1 &= \tfrac{1}{2}h^2 + \tfrac{1}{3}h^3 + \tfrac{1}{6}h^4, \\ G_2 &= \tfrac{1}{3}h^3 + \tfrac{1}{2}h^4 + \tfrac{7}{15}h^5 + \tfrac{2}{9}h^6 + \tfrac{4}{63}h^7. \end{aligned} \quad (126)$$

For rotating state $s = (q, v)$ put

$$\begin{aligned} F_h(s) &= |q| + (\tfrac{3}{2}h^2 + h^3)|q_x| + g_h|v|, \\ D_h &= F_h(z_0) + \sqrt{C(h) G_2}, \\ E_h &= F_h(s_{E0}) + \sqrt{b_E G_2}. \end{aligned} \quad (127)$$

The two algebraic conditions

$$\begin{aligned} E_h + D_h + N_\epsilon G_1 &< \epsilon, \\ \sqrt{b_P} &\geq \sqrt{b_E} + \sqrt{C(h)} + L_\epsilon \sqrt{h}\, D_h \end{aligned} \quad (128)$$

certify nonlinear rendezvous by h. Indeed $\|\Pi\Phi(t)B\| \leq t + t^2 + 2t^3/3$ and $|\Pi\Phi(t)s| \leq F_h(s)$ for $0 \leq t \leq h$. Variation of constants bounds the evader by $E_h + N_\epsilon G_1$ and the relative path by D_h. The first inequality prevents a first exit of either ship. The second bounds the compensation effort. For a physical exclusion radius also require $R_0(1 - \epsilon) \geq r_{\min}$.

Failure of either inequality is inconclusive. Appendix D supplies an inertial half-space certificate for wider separations.

D An inertial half-space rendezvous certificate

Here all quantities are physical and $g(x) = -\mu x/|x|^3$; use $\mu = 1$ for normalized quantities. Put $r_0 = x_{E0} - x_{P0}$, $w_0 = v_{E0} - v_{P0}$ and $s = t/h$. The free-space rendezvous correction and its relative path are

$$a_h(t) = 6r_0/h^2 + 4w_0/h - (12r_0/h^3 + 6w_0/h^2)t, \quad (129)$$

$$r_J(t) = (1 - 3s^2 + 2s^3)r_0 + h(s - 2s^2 + s^3)w_0. \quad (130)$$

Thus $\int |a_h|^2 = C_J(h)$ from Section 8, and direct polynomial integration gives

$$I_J = \int_0^h |r_J|^2 dt = \frac{13h}{35}|r_0|^2 + \frac{11h^2}{105} r_0 \cdot w_0 + \frac{h^3}{105}|w_0|^2. \quad (131)$$

Choose a unit vector n and $r_* > 0$. Require, for every $0 \le t \le h$,

$$\begin{aligned} n \cdot (x_{E0} + t v_{E0}) - \frac{\mu t^2}{2r_*^2} - \sqrt{b_E t^3/3} \\ - \max\{n \cdot r_J(t), 0\} > r_*. \end{aligned} \quad (132)$$

A simpler sufficient scalar check replaces the linear term by the minimum of its two endpoint values and subtracts $\mu h^2/(2r_*^2) + \sqrt{b_E h^3/3} + D_J$, where $D_J = |r_0| + 4h|w_0|/27$. If also

$$\sqrt{b_P} \ge \sqrt{b_E} + \sqrt{C_J(h)} + \frac{2\mu}{r_*^3}\sqrt{I_J}, \quad (133)$$

the response $a_P = a_E + g(x_E) - g(x_P) + a_h$ guarantees full-state rendezvous by h. To prove the corridor, before a first exit the evader gravitational displacement in direction n is bounded below by $-\mu t^2/(2r_*^2)$ and its controlled displacement by $-\sqrt{b_E t^3/3}$. Since $x_P = x_E - r_J$, (132) excludes a first exit of either ship. Their joining segment is in the convex half-space $n \cdot x \ge r_*$, so $|g(x_E) - g(x_P)| \le K_g |r_J|$, $K_g = 2\mu/r_*^3$. Minkowski and the quadrature then prove the resource bound. Choose $r_* \ge r_{\min}$ if an exclusion floor is imposed. A failed sufficient test is inconclusive, and finite-band admissibility is an additional check.

E CW entries and the rational Jensen calculation

Write $S = \sin h$, $C_h = \cos h$, $S_2 = \sin(2h)$. The unit-mean-motion transition matrix is

$$\Phi(h) = \begin{pmatrix} 4 - 3C_h & 0 & S & 2(1 - C_h) \\ 6(S - h) & 1 & -2(1 - C_h) & 4S - 3h \\ 3S & 0 & C_h & 2S \\ -6(1 - C_h) & 0 & -2S & 4C_h - 3 \end{pmatrix}. \quad (134)$$

The symmetric Gramian has upper-triangle entries

$$\begin{aligned} W_{11} &= 13h/2 - 8S + 3S_2/4, \\ W_{12} &= -3(h - S)^2, \\ W_{13} &= 4(1 - C_h) - 3S^2/2, \\ W_{14} &= -11h + 14S - 3S_2/2, \\ W_{22} &= 3h^3 + 14h + 24hC_h - 32S - 3S_2, \\ W_{23} &= 5h + 6hC_h - 8S - 3S_2/2, \\ W_{24} &= 6S^2 + 4(1 - C_h) - 12hS + 9h^2/2, \\ W_{33} &= 5h/2 - 3S_2/4, \\ W_{34} &= 3S^2 + 6(C_h - 1), \\ W_{44} &= 19h + 3S_2 - 24S. \end{aligned}$$

These follow by integrating $\Phi(t)BB^T\Phi(t)^T$ entrywise. For the Jensen example, $y = \Phi(h)\xi = (1, 1 - 3h/2, 0, -3/2)^T$. Evaluate $c = y^T W^{-1} y$ at rational $h = 10, 21/2, 11$ using interval Gaussian elimination. The following construction specifies the interval calculation without requiring an external supplement. For each sine or cosine at argument $x \in \{h, 2h\}$, sum its rational Taylor polynomial through degree 181 or 180, respectively; append the symmetric Lagrange remainder interval $[-|x|^{d+1}/(d+1)!, |x|^{d+1}/(d+1)!]$. Convert fractions outward to 70 decimal digits. Round every subsequent interval arithmetic operation outward, check that each elimination pivot excludes zero, solve $Wq = y$, and form $y^T q$. The deliberately widened enclosures are $|c(h) - \bar{c}(h)| \le 2\,10^{-20}$, with the following rational centers:

h	$\bar{c}(h)$
10	0.08299553059490180337
10.5	0.08164450969532354695
11	0.08014088306972455432

Consequently the Jensen gap differs from $7.630286301036810\,10^{-5}$ by at most $4\,10^{-20}$ and is strictly positive. The argument in Section 2 transfers this leading sign to the nonlinear value for sufficiently small perturbations; it does not supply an explicit largest perturbation.

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